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A Comprehensive Review of Using WSNs and Drones for Improving Crop Production in Precision Agriculture

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ABSTRACT

Precision agriculture (PA) plays an essential role in resource use and crop yields while minimising environmental impact through data-driven farming techniques. The combination of unmanned aerial vehicles (UAVs), the Internet of Things (IoT) and wireless sensor networks (WSNs) has significantly transformed the current state of farming, enabling decisions based on data, predicting outcomes and precise control. This review presents the current developments, challenges and complementary advantages of these technologies to improve agricultural efficiency and sustainability in a comprehensive manner. The search timeframe of this search is 2019–2025. The analysis of the WSN-based systems begins with the analysis of sensing technologies, communication protocols (LoRa, Sigfox, Wi-Fi, Bluetooth, ZigBee, NB-IoT and RFID), sensor architecture, energy consumption and path-loss models, which affect the data transmission in an agricultural setting. It highlights the weaknesses of WSN deployment, such as power consumption and coverage. Second, the use of UAVs in crop monitoring, irrigation, pest detection and resource optimisation is reviewed with references to the incorporation of sensing and data analytics algorithms and the challenges associated with UAV use, such as the short flight duration and energy consumption. Third, IoT-based frameworks are researched in the context of their roles in the PA of real-time monitoring, automated controls and smart decision-making. The findings suggest that a network of WSNs, UAVs and the IoT can be used to enhance monitoring quality, data quality and resource utilisation by multiple orders of magnitude. However, such barriers as energy consumption, connectivity differences, complexity of data and costs will be topical. Overall, the WSN-UAV-IoT combo is a potentially fruitful direction that could assist PA to take a step forward in terms of productivity, sustainability and environmental friendliness.

1 | Introduction

Agriculture is a very crucial factor that promotes food economic stability, security and environmental health worldwide [1]. By 2050, the world is estimated to reach 9.7 billion people, which means that the demand will be high on agricultural products [2]. Sustainable food security depends on the enablement of agriculture. By putting emphasis on the quality and the quantity of the production, it is necessary to ensure that the farmers utilise automated and modernised farming methods so that they can enhance the efficiency and maximisation of the resources used and the volume of harvest [3]. Monitored and automated agricultural systems are crucial for meeting the rising demand for food resulting from climate change and global population growth, all while efficiently managing limited resources. Precision agriculture (PA) is based on continuous monitoring of the soil situation, crop well-being, weather and application of smart irrigation systems [4], enabled through the newest technologies, including wireless sensor network (WSN), drones and Internet of Things (IoT) [5]. PA-based data-driven to enhance productivity and sustainability, as opposed to the conventional farming practices, where the process involves generic manual practices [6].

WSNs are essential in new practice applications, the great pace of technology and other activities. Their increase was very high between 2012 and 2022, with a 0.45 billion growth, now reaching 2 billion [7]. Self-configuration and organisation over the period set by the manufacturing developments are the main advantages of WSNs. Additionally, when appropriate path loss models (PLMs) are applied, their performance is enhanced, which is beneficial in ensuring network reliability, integrity, and capacity [8]. Generally, WSNs are a huge number of battery-powered sensor nodes, which easily emphasises the issues of power and energy usage [7, 9]. The current agricultural networks are based on the use of WSNs in real-time data collection to monitor and regulate the environmental conditions influencing crop production [10]. A WSN is a network of distributed sensors that transmit wireless data collected to a central processing unit, which allows farmers to conduct remote analyses. These networks may be used to improve the performance of the farming industry through the provision of vital data regarding the moisture content of the soil, variation in temperature, humidity levels and nutrient availability [11]. WSN-based agricultural monitoring systems have three essential sensors that are (i) soil sensors that record moisture levels, temperature, pH levels and nutrients [12]; (ii) weather sensors that observe ambient temperatures, humidity and rainfall [13] and (iii) plant health sensors that discern the presence of diseases, pest infestation and vitality of plants through spectral observation [14]. Application of WSN in agriculture saves the expenses of operation as well as facilitating accurate application of resources and minimising the effect on the environment in fields. ZigBee, LoRa and Wi-Fi are examples of wireless technologies that allow the transmission of data gathered by sensors in an efficient manner, allowing monitoring and automation of the farm in real-time [15].

Drones, also referred to as UAVs, have numerous uses, such as rangeland and ecological studies, forestry and agriculture [16]. This application of UAVs or drones offers real-time aerial services and monitoring of activities in modernising the PA operations. UAVs can be used to manage and collect information from small to medium-sized farms in an efficient way. They are characterised by having low environmental effects, high-resolution images at an affordable price, real-time processing of data, high-tech systems, spectral, spatial and temporal imagery, with an image resolution that is better than satellite images [17]. Nevertheless, the price of the UAV is an obstacle to its extensive application as it is considered in terms of sensors, payload, data analytics, flight time, environmental conditions and requirements [18]. Newer drone sensors produce high-resolution maps based on thermal and images, which draw the attention of farmers to problems with crop health, irrigation and disease outbreaks. There are three sensors integrated into agricultural drones: (i) a multispectral sensor that measures reflected data analysis to determine plant health [19], a thermal sensor that measures variations in temperature, which is a potential symptom of water stress and plant diseases [20] and (iii) LiDAR sensor, which develops a complete map of the farmlands and crops in three dimensions [21]. Ground stations are connected by real-time drone information exchange through wireless connections such as 5G, Wi-Fi and RF. The real-time transfer of data also allows farmers to react with high speed as they have improved access to accurate information about crop health and resource management.

IoT has transformed the agriculture sector by connecting sensors and devices in one network. Farm activities can be remotely monitored, automated and controlled by the operators. Farmers can monitor irrigation, fertilisation and pest management without having close supervision. The main IoT applications are as follows: (i) smart irrigation systems, which are based on automated controllers, guided by sensing the level of soil moisture [22], (ii) precision fertilisation uses sensors to identify the nutrient deficiencies and apply fertiliser precisely [23] and (iii) livestock monitoring equipment, which provides wearables with IoT capabilities to track the health, behaviour and location of animals [24]. The implementation of IoT agriculture is based on the use of cloud computing, edge computing, and blockchain technology to produce safe, scalable and data management systems. The combination of LoRaWAN, NB-IoT and cellular networks will provide a reliable data transfer and access to essential information regarding the farm in real-time will be possible [25].

The integration of WSNs, UAVs and IoT technologies transforms PA into a more efficient organisation in its operations, manages its resources and is more environmentally sustainable. WSN sensors utilise low power to detect environmental factors, which include soil moisture, temperature and humidity, and send the data wirelessly. The IoT sensors are used to send the data from the farm to the cloud to be analysed. Cameron drones are used to undertake thermal and multispectral surveys, where they take high-definition pictures to reveal the premature occurrence of crop diseases, pests and nutrient deficiencies. This will enable targeted interventions that reduce the use of pesticides and enhance the yield of harvests. The first-generation IoT automation tools assist farmers in controlling irrigation, fertilisation and pesticide application with the help of real-time data, and consequently, save money on the operation. The association of data with models of natural phenomena, pest behaviour and crop growth will enable planners to better reduce losses. The incorporation of current technology changes conventional farming to data-driven systems that are efficient and sustainable in delivering world food standards.

The novelty of this paper is to examine the concept of WSNs, UAVs and IoT as essential technologies in agriculture and how to employ them in PA. It also addresses the technical issues and the possible areas of investigation to provide the farmers with a thorough understanding of these innovations. The paper compares the progress in the contemporary WSNs, UAVs and IoT by analysing their practical advantages, technological gains and challenges faced in operating within the context of sustainable agricultural improvement and efficiency. It also examines special wireless communication technologies, such as LoRa, Sigfox, Wi-Fi, Bluetooth, Zigbee, narrowband IoT and RFID. It examines WSN platforms, sensors, controllers, measurement methods, power consumption and path-loss models. The previous studies indicate the importance of UAVs in aiding PA by monitoring crops, managing resources and controlling pests, though the issue of flight time and battery consumption is still a concern. This review also discusses the application of IoT in precision farming, which is characterised by real-time monitoring, automatic regulation and system choice. The last issue of the present research is the prospects of PA development.

Figure 1 shows the example architecture of a WSN, UAV and an IoT system customised to be used in agriculture in four steps. The sensor nodes are initially installed in the farm to detect the environmental factors such as rainfall, soil moisture, light intensity, air temperature and humidity. The information obtained is transmitted to a receiver through a router node, namely, a LoRa node attached to a UAV. When the data are received, it is stored in the cloud, which helps in training the model and also in decision-making related to irrigation, estimation of crop yield, farm management and also in fertilisation. LoRa is the suggested technology of communication in this arrangement. The contributions of this paper can be summarised as follows.

- This review provides a critical examination of systems-based WSNs, UAVs and IoT technologies to use in PA to have a comprehensive idea of their applications.
- This paper classifies PA technologies that are a combination of WSN, UAV and IoT systems. The classification offers scholars and agricultural practitioners valuable information and new opportunities for agricultural innovation in the future.
- Different agricultural studies that involve the use of WSN and UAV technology are compared with a focus on performance metrics. Such a comparison assists the researchers in choosing the most appropriate technologies to be used in their research and in their practical uses.
- This study highlights the high challenges and drawbacks of the WSN and UAV technologies in the agricultural sector. It sheds light on the possibilities to improve the usability of the system and overcome the existing problems.
- Several PLM analyses, which are frequently applied in agricultural fields, aim to improve the reliability, integrity and capacity of WSNs.
- The future trends and possible future applications in the field of agriculture are discussed, which provides the researchers with some innovative ideas as well as gives them a lot of background to explore the subject.

The review paper is structured as follows: Section 2 presents the motivation behind this study. Section 3 explains the methodology of this study. Section 4 displays the types of sensors used in agriculture applications. Section 5 outlines wireless communications, reviews prior applications in agriculture, compares them and presents path-loss models and challenges in WSN. Section 6 presents previous agricultural studies using UAVs/drones, with comparisons and challenges. Section 7 introduces the discussion of the latest work-based WSNs and drones. Section 8 highlights the use of IoT in agriculture. Section 9 demonstrated the common challenges and limitations of WSNs, UAVs and IoT in agriculture. Section 10 highlights the future works. Finally, Section 11 concludes the paper.

2 | Motivation Behind This Study

The necessity to introduce a solution to the problem of world food security as soon as possible prompts the reevaluation of WSNs and drones to enhance crop production in the

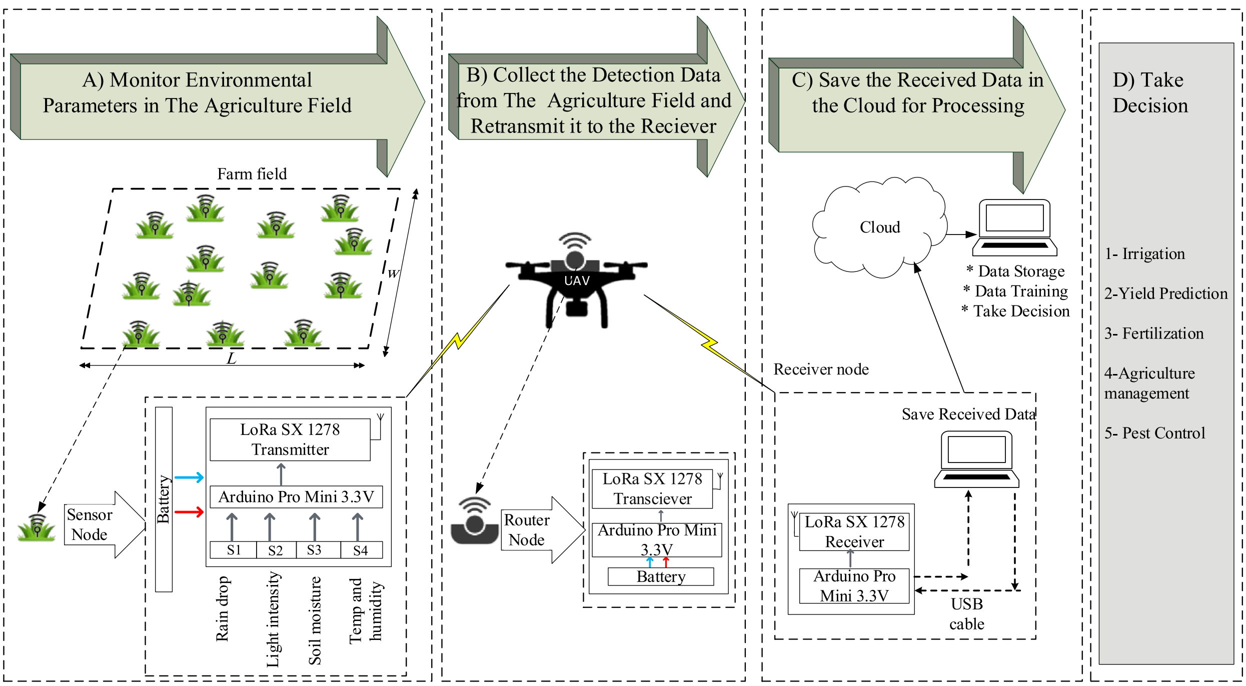

FIGURE 1 | Proposed example of WSN and IoT system architecture for agricultural applications.

agricultural industry. With the increasing population in the world, there is a need to make agricultural systems more efficient, sustainable and resilient to produce more food that will sustain the growing demand for food in the world. Conventional farming approaches do not usually achieve high efficiency in resource utilisation and are inadequate to deal with climate change, pests and illness, which is why there is a pressing necessity to find novel solutions. WSNs and drones are a revolutionary technology because they can deliver up-to-date data, insights and automation that can benefit contemporary farming methods. The tools improve environmental management through improved monitoring, enable fast and data-driven decisions and more efficient crop management. The application of these technologies helps farmers to increase yields and sustainability and minimise the waste of water, fertilisers and energy. This paper aims to synthesise the current research, finding the gaps and discussing the potential of these technologies to be combined to define the future of smart agriculture. In this field, data communication is dependent on PLM. The improved knowledge of the signal propagation in agriculture assists in bridging the disparity between innovative wireless technology and implementation in the agricultural field, facilitating more productivity and sustainability. The scientific study on drones points to how they can transform the agricultural sector by enhancing precision, minimising environmental impacts and improving farming operations such as crop surveillance, pesticide spraying and irrigation. This paper aims to bridge the gap between industrial solutions and traditional farming to encourage improvements in the fields.

3 | Methods Followed in This Study

The research approach of this paper is that of a systematic review of existing wireless technologies and techniques in the agricultural environment. The study was conducted between June 2024 and March 2025, during which a systematic selection of articles was made by methodological criteria. The searches in large scientific databases, such as ScienceDirect, Scopus and Google Scholar, were limited to publications from 2019 to 2025. In Google Scholar, the search query was 'WSNs' AND 'agriculture', which provides 14,700 results, with the addition of 'power consumption' giving 9900. 'UAVs' AND 'agriculture' searches yielded 20,600 articles, and 'IoT' AND 'agriculture' yielded 196,000 articles. The chosen papers, based on their relevance to the themes and the number of citations, corrected by the publication year, are used to prioritise the similar works. Papers of good quality on less-discussed research areas were also added, with or without citations. Having examined such studies, we were able to compile and analyse the academic literature and case studies that comprehensively cover the use of WSNs, drones and IoT in PA applications, discussed in terms of design, functionalities and performance of each technology in real and virtual terms. The work of these technologies was evaluated taking into account the paramount parameters such as the type of controller, sensors deployed, goals, methodology, measurement parameters, power consumption, size of the farm, the type of drone, flight time, battery capacity and the type of agricultural application. Their strengths, which are enhanced productivity or reduced labour costs, are also noted in the analysis. Past research has cited drawbacks of WSNs and drones in farming, such as initial expenses, technical difficulty and energy utilisation. Through identifying such gaps, the research will seek to investigate how these technologies can be modified or integrated to come up with holistic solutions. This systematic review offers a better insight into the way these tools can be refined or combined to encourage sustainable and efficient agricultural practices.

4 | Sensors in Agriculture

Sensors play an important role in PA since they enable real-time gathering of data and automated farming. Different sensors, including optical sensors to measure plant health and insect infestations, soil moisture sensors to manage irrigation, temperature and humidity sensors to control the climate of greenhouses, electrochemical sensors to evaluate the nutrients and pH levels of the soil and so on, help farmers make a well-informed choice. Moreover, remote sensing devices, such as drones, IoT and satellites, can give the full picture of the state of crops. WSN enhances efficiency since it transmits real-time information to do automated watering, fertilisation and pest control. Sensors play a significant role in enabling durable and smart agricultural practices by enhancing resource management, reducing waste and increasing yields. WSNs are crucial for enhancing food production across various agricultural applications, including soil moisture monitoring, irrigation, weed management, storage control, yield prediction, fertiliser optimisation and control, early-stage management of crop diseases and pests and energy conservation [26]. Table 1 provides details on the most widely used sensors in agriculture, including their operational ranges, the physical phenomena they detect and their sensitivity (accuracy and resolution). It is worth mentioning that all sensor information details are taken from (https://www.alldatasheet.com/).

5 | Wireless Communications in Agriculture

Wireless communication in agriculture enables efficient data sharing among equipment, systems and stakeholders across the field. It supports precision agriculture by allowing drones, sensors and devices to monitor soil, crop health and environmental conditions in real time. Bluetooth, Wi-Fi, cellular networks, Sigfox and LoRa are technologies that guarantee effective communication in agriculture. This information assists farmers in making better decisions, managing resources more effectively, maximising yields and reducing wastage. Wireless surveillance lowers manpower and time wasted in performing manual surveillance. Irrigation and pest control systems can be controlled wirelessly in order to be more precise. Moreover, predictive analytics, based on wireless technology through IoT and cloud technologies, are applied to increase planning and risk management. With the help of GPS, it allows mapping autonomous machines, such as harvesters and tractors, and navigation is accurate. These programmes help address sustainability challenges by reducing overuse. Ultimately, wireless communication revolutionises agriculture by enhancing efficiency, sustainability and data-driven capabilities. Different

TABLE 1 | Common types of sensors used in the agriculture sector.

Type of sensors	Operating range	Sensed physical phenomena	Sensitivity: Accuracy	Sensitivity: Resolution
1- Soil sensors [27]				
Hydra probe II soil sensor	−10°C to +60°C, 0%–100%	Soil moisture for inorganic and mineral soil	± 0.1 water volume fraction	0.1%
MP406	−50°C to +1500°C, 0%–100%	Soil temperature and soil moisture	± 1% volumetric water content	0.01%
Pogo portable soil sensor	−10° C to +65°C, 0%–100%	Soil moisture for inorganic and mineral soil	> ± 0.03 water volume fraction and ± 0.1°C	N/A
ECH2OEC-5	−40°C to +60°C, 0%–100%	Soil temperature, soil moisture and conductivity	0.03 m^3/m^3 volumetric water content	0.01% in mineral soils and 0.25% in growing media
WET-2	−5°C–50°C, 0%–100%	Water of soil content, electrical conductivity and the temperature of the soil	3%	10%
VH-400	−40°C–85°C	Soil water content.	2%	N/A
107-LtemperatureSensor	−35°C to +50°C	Air/soil/water temperature	± 0.4 to ± 0.9	N/A
2- Crop sensors [28]				
Leaf wetness sensor	−40°C–85°C	Leaf wetness	5%	0.1%
LT-2M (leaf temperature)	5°C to +50°C	Absolute temperature of leaves	< ± 0.08°C	15:1
TPS-2 portable photosynthesis	−10° C–60°C	CO_2, H_2O and light	± 0.3 at 25°C	1 ppm
PTM-48A photosynthesis monitor	0%–100% RH, 0°C–50°C	Carbon dioxide and water vapour exchange in leaves	± 0.02 LPM	11 bit
Cl-340hand-held photosynthesis	0°C–45°C	CO_2/H_2O, pressure and leaf temperature Measures chlorophyll fluorescence and photosynthesis simultaneously	< ± 0.02, ± 2% at 10% RH. ± 3.5% at 90% RH, 5 mol m^{-2} s^{-1}, ± 0.1°C and ± 0.3°C	0.1 ppm
YSI 6025 chlorophyll sensor	~0–400 µg/L, 0 to 100 RFU	Chlorophyll	N/A	0.1 µg/L Chl 0.1% RFU
Field scout CM1000TM	12–72 inches	Plant health by calculating the normalised difference vegetation index (NDVI)	± 0.05 measurements	Higher light levels enable greater resolution

(Continues)

TABLE 1 | (Continued)

Type of sensors	Operating range	Sensed physical phenomena	Sensitivity	
			Accuracy	Resolution
3- Environmental sensors [29]				
Met station one	0–50 m/s, 0°–360°.	Speed and direction of the wind	± 2% and ± 5°	0–50 m/s and 1°
SenseH2hydrogen sensor	−20°C–80°C, 5%–95% R.H	Hydrogen monitoring	0.25%–4.0%	0.5%
CM-100 compact weather station	0°–360°, 0%–100%, 0–50 m/s, 0°–360° and 600 to 1100 hPa	Air temperature, relative humidity, wind speed, wind direction and barometric pressure	−50°C to +50°C, ± 3%, ± 0.5 m/s or 5%, ± 5° and ± 0.35 hPa	0.1°C, 1.0%,0.1 m/s, 1.0° and 0.1 hPa
CS300-LPyranometer	−40°C to +70°C, 0%–100%	Total sun and sky solar radiation for solar, agricultural, meteorological and hydrological applications	± 5%	0.2 mV/W/m^{-2}
Tipping bucket rain gauge	0°C–70°, −20°C to +70°C	General rainfall monitoring	< ± 2%	0.1–0.2 or 0.5 mm/tip
ECRN-100high REC rain gauge	0°C–60°C	General rainfall monitoring	0.2 mm	1 tip
RG13/RG13H	1°C to +85°C	Rainfall/liquid precipitation	2%	0.2 mm
HMP45C	−40°C–60°C	Air temperature and humidity	0.2°C–0.5°C and ± 2% to ± 3%	0.033°C and 1%
SHT71/SHT75/SHT11	0°C to +49°C	Humidity and temperature	± 3%, ± 0.4°C	0.03% and 0.01°C
Cl-340	0°C–45°C	CO_2, H_2O, temperature and light as well as measure chlorophyll fluorescence and photosynthesis rates simultaneously	< ± 2% up to 2000 ppm, ± 2% at 10% RH, ± 3.5% at 95% RH, ± 5 µmol 0–2500 µmol/m^2/s, ±0.1°C and ± 0.3°C	0.1 ppm, 0.1%,
Met one series 380 rain gauge	−50°C to +50°C	Rain and/or snow in all environments	± 0.5% at 0.5″/hr and ± 1″–3″/hr	0.1 mm per tip
WXT520 compact weather station	−52°C to +60°C	Air pressure, humidity, temperature, wind speed, wind direction and rainfall	± 0.5 hPa to ± 1 hPa, ± 3% RH to ± 5% RH, ± 0.3°C, ± 0.3 m s^{-1} or ± 3%, ± 3° and ± 5%	0.1 hPa, 0.1% RH, 0.1°C, ± 3° and ± 5%
RM Young (model 5103)	−50°C to 50°C	Wind speed	± 0.3 m/s	0.5°

wireless communication technologies for the smart agriculture spectrum have been presented to improve agricultural yields [28, 30].

A standard of communication protocols with low power that is used for an agricultural domain can be classified into WSNs, low power wide area Networks (LPWAN), cellular networks, short-range communication, satellite, radio frequency (RF) and infrared (IR), depending on the range of communication fields [31]. These wireless technologies are highlighted in Figure 2.

5.1 | Previous Works-Based WSN in Agriculture

5.1.1 | Long Range Radio (LoRa/LoRaWAN)

LoRa is a wireless communication protocol stack for low-power indoor transmission. On top of LoRa, LoRaWAN is a LoRa wide area network protocol developed in 2015 that enables long-range communication over 15–20 km and 2–8 km in rural and urban areas, respectively [32]. It operates on a low supply voltage of 2.7–3.7 V [33]. The LoRaWAN network has a star topology consisting of end devices (EDs), gateways (GWs) and the LoRaWAN network server (NS). The ED uses LoRa to communicate with the GW, whereas the latter uses an IP protocol to communicate with NS [34].

Yang et al. [35] proposed a low-power intelligent agricultural monitoring system based on LoRa wireless technology to monitor and control greenhouse agriculture. Different sensors, including humidity, light intensity, soil temperature and carbon dioxide concentration, were used to measure the required parameters for light control, cooling control, man–machine control and irrigation control. The intelligent system was tested in the laboratory and deployed in a dragon fruit greenhouse, where data were collected and transmitted via LoRa to the server. The application platform can access the data and receive commands from it. Laboratory testing and actual experiments improved system feasibility and helped farmers manage growing crops efficiently in real time with a sleep current of 0.3 μA. However, this system was limited to only the greenhouse.

Codeluppi et al. [36] presented a low-cost IoT-oriented platform based on LoRaWAN for smart farming processes, named LoRa farm, to achieve and optimise a more sustainable agricultural environment. The proposed platform consists of two paradigms, one deployed in a greenhouse of $20 \times 9 \times 5$ m^3 named the greenhouse module and another deployed in a vineyard farm referred to as the vineyards module, which was based on inner nodes and end nodes, respectively, for gathering environmental data, aiming to improve the generic farm's management in an efficient customisable way. LoRa collected the data samples periodically for 10 min, except that the temperature and humidity sensor in the vineyard field data were regularly collected for 30 min, saved then sent to the LoRaWAN-GW, which forwarded it to the network server. The mismatched data-collection intervals were used to evaluate throughput and enable the deployment of many sensors to maintain network traffic. The sensor nodes of the LoRa farm platform, on outdoor and indoor processing boards, enabled a wake-up and deep sleep (at least 98.5%) operational pattern. They are powered by 3.7 V LiPo batteries with 1800 mA and recharged by a 1 W solar panel measuring 80×100 m^2, which fully charges in 13–14 h. Accordingly, the validation campaign lasted 3 months. It resulted in a throughput rate of 80% or higher, depending on the data collection interval settings of 10 and 30, with an increase in the sensor nodes' lifetime from 9 to 14 h, where the current consumption for inner nodes/end nodes was approximately 9/0.9 mAh, respectively. These considerations have been practically validated by investigating data forwarding over 81 days with no additional impact and by the battery recharging even in adverse weather conditions. However, some samples were lost during rainy days, so the measurements were unreliable.

Prasad et al. [37] presented an intelligent monitoring and management system for agricultural applications in rural areas, using a LoRa-based WSN to supersede conventional agrarian approaches. It aimed to simultaneously collect parameters in the agricultural field, such as humidity, temperature and soil moisture, and send them to the cloud without using routers to address communication failures and conserve transmission energy. LoRa technology gathers data from sensors deployed in agriculture and processes it with Arduino. The resulting data are transmitted to the LoRa gateway, which then communicates with the cloud server via Ethernet or WiFi. Additionally, Arduino sends control commands to operate motors when data falls within optimal ranges. This system demonstrated high accuracy in parameter measurement, covering distances up to

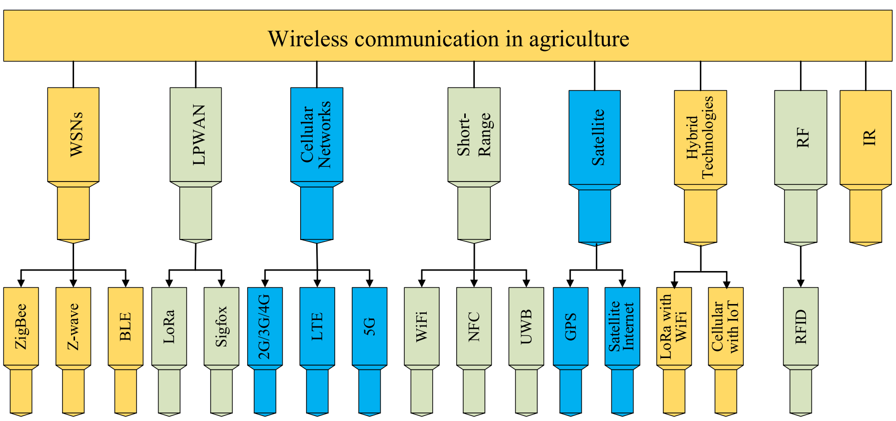

FIGURE 2 | Wireless communication technologies in agriculture.

10 km while consuming only 66 mW per hour. However, its simulation capabilities remain limited.

López et al. [38] developed a sustainable fuel cell system powered by forage grass to feed the sensing edge of a long-range IoT soil-monitoring system based on a LoRa network. A stenotaphrum secundatum grass-type in a 0.015 m^3 chamber with carbon-plate electrodes was used to generate power, and the power output was quantified as the number of transmitted LoRa packets. A low-cost humidity and temperature sensor was used to collect data via LoRa and send it to the cloud via LoRaWAN. An integrated circuit was used to harvest energy for a 4F supercapacitor. The state transition graph was used as a power management strategy to conserve energy in the edge-computing sensing node. This development resulted in a maximum power generation of 1.8 mW, a power consumption of 1.51 mW and 18 consecutive transmissions by the LoRa network without a drop. Although the power generation capacity seems stable, the soil water content decreased, leading to reduced power generation for several days.

5.1.2 | Sigfox

Sigfox is an LPWAN technology module focused on low-data-rate transmissions, such as machine-to-machine applications and sensors. The frequency bands used were 868 MHz (Europe) and 433 MHz (Asia)—with a maximum payload of 12 bytes per uplink [39]. Sigfox has an extensive distance coverage of 10–40 km [40].

Ahumada et al. [41] presented an irrigation system management-based Sigfox and microcontroller ESP32-LoRa for monitoring and acting on environmental parameters. The system provides local communication and access to IoT platforms using the ESP32-LoRa and Sigfox microcontrollers, respectively. It was implemented in an olive orchard of 37.66 ha, HELTEC Wi-Fi LoRa 32 board placed at a height of 1.2 m to collect data from moisture sensor nodes and send it to the GW every 15 min, then the power was interrupted by using a TPL5110 timer module to reduce the consuming current to 25 μA during the interruption time. As a lot, another HELTEC Wi-Fi LoRa 32 board was used on the hydrant valve, which is connected to the GW in the same period above, to check priority and time settings. Finally, the data are sent from the local GW to the IoT platform via the Sigfox web platform. Then, the data were analysed, and the time settings for each hydrant node were generated and executed in priority order. Regarding the mention, the system was powered by two lithium-ion cells with a nominal capacity of 3000 mAh. The distribution of the proposed system resulted in high stability and robustness, providing a sensor node lifetime of more than 5 years. However, automation in rural areas has faced difficulties due to energy availability and connectivity.

PITU et al. [42] presented an agricultural system based on Sigfox technology that monitors, collects and transmits environmental variables related to optimal plant growth. Sigfox technology is integrated with a robust IoT platform application, including a shielded sensor that collects light, barometric pressure, humidity and temperature as well as a three-axis accelerometer, reducing human labour and increasing productivity by monitoring the environment. The system is programmed in Python and tested to measure climate parameters. Accordingly, the system will reduce resource usage, minimising current consumption to 15 mA at input voltages of 3.3 and 5.5 V. However, the system was costly and insecure for small and medium farmers.

Yamazaki et al. [43] established an IoT-based intelligent agriculture system based on sensor nodes developed for Sigfox communication, using open-source general-purpose devices. Sigfox has been explicitly focused on regulating the number of data transactions per day for users using a numerical model, thereby increasing battery lifetime. Eight of the proposed sensor nodes have been placed in a strawberry house with a height of 1.8 m to investigate the plane measurement. On the other hand, 10 Sigfox base stations constitute the infrastructure for bidirectional data transmission over a distance of 26.35 km, requiring space diversity to send the received messages to the Sigfox cloud via the Internet Protocol. The Sigfox cloud was used to reconstruct messages and avoid duplication. Then, the messages were resent to the application server via the Hypertext Transfer Protocol. The path would depend on the uplink and downlink. In contrast, the system consumed 0.02 mA/day in sleep current, reducing the original consumption by 1/40 by increasing the number of batteries, increasing sleep time via minimising transactions and reducing the payload size, which the user adjusted according to an unpublished specific operator algorithm. However, the system was limited to simulation.

5.1.3 | Narrowband IoT (NB-IoT)

The narrowband IoT is a wireless communication technology designed explicitly for IoT and cellular devices [44]. It is characterised by the objectivity of scaling WSN devices to be more dependable and usable, using the long-term evolution licence [8]. It requires constant synchronisation, which consumes more battery power than LoRaWAN, which is well-suited for sleep/wake applications. Additionally, the peak current required for continuous linear transmission of frequency division multiple access or orthogonal frequency-division multiplexing is widely needed [34]. On the other hand, NB-IoT networks require high data rates and low latency, making them suitable for WSN applications that are intolerant of delay and require high data rates [45].

Xiaokang et al. [46] presented a farmland irrigation system based on NB-IoT as a wireless communication technology with low power consumption. NB-IoT wireless communication transmits collected data from the farmland to a personal computer, enabling real-time control of an intelligent water valve via a software system. To assess the system's accuracy and reliability, 12 sensor nodes were arranged in an equilateral shape in a general packet radio service deployment, centred on an IoT gateway. The network sends the collected data to the upper computer for analysis. Setting an appropriate moisture content threshold enables complete control of the intelligent water valve in farmland in PA, increasing crop yields, saving water and creating a suitable growing environment. However, the NB-IoT module's performance is based on the power supply design.

Valence et al. [47] presented a test of NB-IoT in the agricultural field, with a fundamental system architecture, based on actual rural-area trials. A comparison of network performance with the general packet radio service standard was presented to assess the feasibility of applying the NB-IoT industrial system in agriculture. The received signal strength indicator value and packet loss ratio were based on evaluating NB-IoT and general packet radio service performance—the test extended coverage for sensors installed in fields with high penetration losses. However, the NB-IoT protocol has a delay-tolerance property, making it unsuitable for real-time applications.

Tsakalidi et al. [48] presented a novel prototype of a quasi-smart NB-IoT-based water management platform. Since irrigation is directly associated with soil moisture, soil hydration status was regularly and accurately tracked and reported to a centralized entity for processing. To develop an AREThOU5A IoT platform using NB-IoT for data communication, an experiment was conducted in an irrigated vineyard in Peta village with 11 acres and 3300 grape stumps. Battery-powered soil moisture and temperature sensors were used to collect data at depth and transmit it to AREThOU5A IoT end nodes via NB-IoT. Then, the measured data are sent to the cloud using the Constrained Application Protocol/Internet Protocol version protocol over a public mobile network. The data maintained on a cloud back-end can be accessed via an application programming interface from any Internet connection in real time. The proposed system provided numerous novel features regarding operating solutions, hardware integration, networking and connectivity support. However, the prototype provided low data speeds or low transmission rates.

5.1.4 | Bluetooth

Bluetooth, a wireless communication technology, has two categories: classic Bluetooth and Bluetooth Low Energy (BLE). Its concepts are discovery, establishment and management of connections [49]. BLE differs from classic Bluetooth, requires less power, is simpler and is cheaper [50]. Therefore, the latter is concerned with applications that operate at low data rates and low duty cycles. Furthermore, BLE can enter sleep mode when there is no activity, and its power consumption is only 3% that of Wi-Fi. Therefore, power and cost are more efficient than classic Bluetooth [51]. BLE is a network-hopping technique that uses any available channel from a pool of 79, reducing interference [52]. In 2017, the BLE mesh profile was introduced, enabling the connectionless operation of BLE multihop networks [53].

Kirci et al. [54] presented an innovative greenhouse prototype tailored to local climatic conditions. Sensors collected the data to improve the project, which aimed to question the possibility of different plants growing in the same greenhouse. Other sensors were used to collect data for the innovative greenhouse prototype, which was developed using software running on an Arduino Uno microcontroller board programmed in C for automatic control. In addition, Arduino was programmed for manual control by developing a software application in App Inventor for Android to monitor, analyse and evaluate data collected from the greenhouse. Bluetooth was used to communicate with the innovative greenhouse using Android-based mobile devices. The prototype could be continuously controlled for 7 days a week, operating both autonomously and manually; however, some tested plants grew faster in pots.

Rosa et al. [55] presented a wireless sensor platform for temperature, humidity and light detection that could be battery-free and self-powered. To reduce the maintenance intervention required after battery replacement, plant growth and health parameters in the cultivated area can be easily monitored via the installed sensors. The system consists of a battery-powered remote base station and a platform implemented with off-the-shelf solar-harvesting energy components on a printed-circuit board measuring 2×2 cm^2 and 0.45 cm thick, respectively, placed directly on plants' leaves. The device platform can harvest energy, collect vital parameters and communicate via BLE with the BS. Experimental measurements between the sensor platform and a remote BS were conducted in rural areas, achieving a reliable communication distance of 160 m with RSSI dropping below −96 dBm. However, the system has limited greenhouse applications.

Wang et al. [56] introduced a wireless sensor system powered by low-energy solar technology that measures temperature and humidity via Bluetooth Low Energy. Designed to reduce data transmission bottlenecks in real-world settings, it consists of four main components: a sensor module, a power supply, a BLE module and a data display unit. The sensor module gathers data and transmits it in real time via the BLE module to a receiver equipped with a BLE 4.2 USB Dongle. Users can view all data on a mobile app or a computer, allowing effective system monitoring. Although suitable for greenhouse environments, this system is not intended for large-scale deployment.

Raina et al. [57] developed a power-saving PA platform based on BLE to monitor soil moisture. The study compared the power consumption of sensor nodes and the gateway, an area previously unexplored. Each soil moisture sensor, powered by a 620 mAh/3 V battery, lasted 114.18 h, whereas the gateway, which also functions as a monitoring device, lasted 106.74 h with a 15,600 mAh/4.2 V battery. The receiver included a low-noise amplifier to reduce packet loss and extend the communication range. The gateway was equipped with sensors for light intensity, wind speed (anemometer) and temperature and humidity, transmitting data to the cloud via GSM. The average errors across measurements were 0.1 for the soil moisture, 1.9 for light intensity and approximately 1.33 and 1.42 for temperature/humidity and wind speed, respectively.

5.1.5 | ZigBee

ZigBee is a short-range communication standard used in various applications and based on personal area networks [58]. ZigBee performance depends on several factors, including deployment setup, data encryption, transmission power, baud rate, communication distance and the number of hops. These factors significantly influence key metrics, including power consumption, latency, link quality and throughput [59]. ZigBee is widely used in agricultural applications, such as irrigation systems [60], crop yield [61] and smart greenhouses [62].

Sharma et al. [63] presented an innovative WSN-based agricultural system for monitoring, integrated with solar energy harvesting, to maximise network lifetime. Twenty solar-harvesting energy WSN nodes were deployed at a distance/area of 20×50 m^2 and a single GW. On the other hand, sensor nodes deployed without an energy harvester were similarly deployed to improve the proposed solar-harvesting WSN model. The farm was monitored, and sensor nodes collected data. Then, the collected data are sent to ZigBee, which forwards it to the IoT cloud. At the same time, the user accepts such data and can make command decisions remotely. It is worth noting that the harvester circuit consists of small solar panels, a rechargeable battery (a single lithium-ion cell at 3.60 V with 600–850 mAh) and a DC-DC converter, and the system operated at a duty cycle of 25%. The system was experimented with practically. The results showed a prolonged network lifetime of about 115.75 days with a throughput of 160 K bits/s, whereas the network without a harvester circuit decreased the battery lifetime by 5.75 days and reduced the throughput to 100 K bits/s. In contrast, the collision has increased by about 29.90 times.

Agrawal et al. [64] presented an IoT system for PA using WSN technology, estimating energy requirements and improving the duty-cycling algorithm. The energy requirements of the ON and OFF intervals for the sensor node's transmitting messages and the data aggregation algorithm were applied to improve duty cycling. To evaluate the number of ON periods, it was assumed that the base station, a constrained-energy device connected to a solar panel, operates in three power modes at random times: high, medium and low, represented as 1, 2 and 3, respectively. The product density model is the first-order deviation from the ON probability function. The last is used to estimate energy requirements at the base station. Duty cycling was improved by computing the number of hops to the base station to select the shortest path, with all paths stored in a table at the base station. Accordingly, addressing the random energy requirements at the base station was investigated, achieving 6%–7% energy savings and a significant improvement in network throughput under different weather conditions. However, the experiment was limited to simulation.

Sadowskia et al. [65] compared power consumption in an IoT agricultural monitoring system using WSN technologies: IEEE 802.11g (WiFi 2.4 GHz), LoRaWAN and ZigBee. The system can harvest energy and use these different technologies to determine which has the lowest power consumption and prolongs the network lifetime. The sensor data were sent to the base station via different wireless technologies for four trials. This resulted in current consumption of 171.17 mA, 69.36 mA and 29.33 mA for WiFi, ZigBee and LoRaWAN. With a 6600 mAh battery, the first trial was without a solar panel, resulting in 29.06 h, 80.28 h and, after depletion, 166.23 h, 88.25 h, 189.22 h and 28.85 h for WiFi, ZigBee and LoRaWAN, respectively. At the same time, the throughput and transmission range were 54 Mbit/s, 250 kbit/s, 50 kbit/s, 50 m, 120 m and 15,000 m, respectively, for WiFi, ZigBee and LoRaWAN. Accordingly, LoRaWAN was the most suitable for the most extended period, followed by ZigBee and WiFi. However, WiFi consumed the highest power consumption but had much higher throughput.

Lozoya et al. [66] presented an irrigation closed-loop control system based on WSN with an energy-efficient communication strategy. Three soil moisture sensors are buried in an effective root zone of 60 cm in each of four divided pecan field areas, each of 25×100 m^2, with a distance between each sensor of 3m, which are used to measure the water contents dynamically to identify and validate the specific process of irrigation and based on ZigBee as a wireless communication protocol. Furthermore, a self-triggered control algorithm reduced communication traffic by controlling the irrigation process, thereby minimising power consumption. Sensor nodes are woken to collect soil moisture sensor data and transmit the average, along with weather data collected by a weather node. To control the prediction, the subsequent sampling goes to sleep according to the sleep/wake algorithm. The self-triggered algorithm controls the actuator's role in the irrigation process, reducing the number of communication messages. Accordingly, the results showed that communication messages reduced by more than 85% yielded power consumption reductions of up to 20%, with efficient irrigation and reduced water consumption. However, this system had high irrigation errors.

Morais et al. [67] presented a robust, flexible and low-cost wireless device for data acquisition that can be deployed in various applications powered by solar energy for wide-range PA. Multiple wireless communication models, including ZigBee-based XBee, were used to create a WSN that acted as coordinators and provided mesh network capability. LoRa/LoRaWAN provided directivity for transmitting data packages to GWs located 15 km away. In addition, GSM/general packet radio service technology has been used to enable high-speed high-data-rate connectivity between GWs. The emerged wireless technology devices, named (SPWAS'21 devices), were tested in a vineyard to collect agro-meteorological data every 15 min, forming a mesh network of four devices via a ZigBee link to transmit and relay data to the WSN coordinator and GWs. They added the five one-operated over the LoRaWAN link to connect to the wave system using a general packet radio service 2G connection. The average power consumption was approximately 140 μA in active mode and 83 μA in sleep mode, daily. However, mesh networks require routers that rival general packet radio service modems in power consumption, so they consume more power for a fraction of the time.

5.1.6 | Z-Wave

Z-Wave wireless technology is a mesh network protocol originally introduced for indoor automation [68]. Z-Wave devices, such as ZigBee, can be used for several monitoring and control systems with minimal interference because they operate at 908.42 MHz (US) and 868.42 MHz (Europe), not the congested 2.4 GHz spectrum [69]. However, Z-Wave provides limited coverage [70], is unsuitable for low-data-transfer applications and has a single point of failure, resulting in low reliability [53]. In addition, Z-Wave is an efficient solution for small farmhouses in remote areas with limited network connectivity [71].

Effah et al. [72] presented a case study about designing an energy-efficient, adaptive, self-healing, stable, autonomous, affordable and cluster-based WSN-specific Agri-IoT framework. To ensure network optimisation, a taxonomy of multiobjective optimisation metrics can deliver improved network performance. The Agri-IoT framework consisted of sensor nodes, base stations, actuator controllers and a gateway. At the same time, the sensor nodes consisted of a sensing unit, power unit, control unit and communication unit, which used low-power short-range standards, such as BLE, LoRa, Z-Wave, WiFi, or ZigBee, for on-field communication, depending on application requirements and cellular technology. Sensor nodes deployed in an agricultural field collected the data. These data are transmitted via the communication unit to the gateway and then sent to the cloud via cellular technology. Data maintained can be accessed through the web application. This tutorial introduced new taxonomies of architectural layers, faults and multi-objective optimisation metrics for cluster-based Agri-IoT networks as well as a three-tier objective framework with standard measures for the design of an efficient supervisory protocol. However, the framework must meet expectations for stable performance guarantees, and there were no practical results or an in-depth contextualised overview of Agri-IoT.

5.1.7 | RFID

RFID is a standard technology for tracing food, animals, goods, flows and assets in a logistics system. RFID sensing technology has emerged in agriculture and established a new series of applications [73]. In ref. [74], a UHF RFID-based system for measuring relative humidity, soil moisture, light and temperature was presented. A unique passive compact UHF RFID tag is based on wirelessly hosting temperature sensors for subsoil measurements [75]. Meanwhile, ref. [76] presented a merging specific ground with aerial-based technologies (small unmanned aircraft system (sUAS) with RFID) as a comprehensive approach to enable on-demand automation for plant inventory. In ref. [73], RFID was used to track animals and food. In ref. [77], a smart RFID tag was used to measure soil parameters critical to crop development, including oxygen concentration, temperature and humidity.

Rennane et al. [78] presented the development of a low-power RFID sensor device for crop growth management and equipment maintenance in modern greenhouse applications. An enhancement to the EPC global C3 SL900A sensor tag was presented by integrating multisensing capabilities and energy harvesting to provide an efficient solution for modernising greenhouses. The RFID Reader of Impinj Speedway Gen2 with a horn antenna was used, and the sensor data were transferred over the serial-peripheral interface bus through an external microcontroller unit MCU for direct access. By integrating this chip's rectifier circuit, a maximum voltage VRectf of 3.4 V and a maximum current of 200 A can be provided, and the output can serve as an energy source when converting the received RF power to DC power. However, the measurements indicated a maximum efficiency at a distance of 1.65 m, with an output voltage of 1.87 V, indicating a low relative efficiency (≈12%).

Savitha et al. [79] presented a hardware implementation for warehouse monitoring in a greenhouse. The diverse agricultural sector and vast population led to food security challenges, so integrating greenhouses and warehouses improved the sustainability and quality of food production. The warehouse system consists of an RC522 RFID reader, a flame sensor and IR sensors, all controlled by an ESP32 to provide monitoring and access control functionality. In contrast, the RC522 RFID reader reads users' RFID tag identifiers to activate the automatic door-opening mechanism and sends a notification to a central server or mobile application for indication. This helped with remote monitoring via a web application. However, integration with blockchain technology is required.

Hattaraki et al. [80] presented a photovoltaic module that powered an Arduino UNO equipped with a fire sensor, an LDR, a soil moisture sensor, a temperature sensor and a humidity sensor. This research aimed to integrate IoT and solar power to regulate and monitor a greenhouse, enabling farmers to maintain optimal environmental conditions in real time. RFID tags and readers were used to grant access to the greenhouse, prevent unauthorised entry and assist farmers in maintaining overall control and monitoring the greenhouse system. However, additional work is required to grow different plants in the greenhouse.

5.1.8 | WiFi

This wireless is very popular in the miniature appliances such as tablets, laptops and desktops. The WiFi possesses limited data communication range [81], where its transmission range is approximately 20 m in indoor and 100 m in outdoor setting [82]. The WiFi is an extension of the PA framework architecture that connects various types of systems through ad hoc networks [83]. WiFi can be compared to mobile phone applications in agriculture, where service access is transmitted over distance with a short message sent to monitor and control the protected crops. However, it is not popular in agriculture since it involves constant listening that consumes large amounts of power. Besides, WiFi is slow with big data transfers and used in applications that require multihops [84].

Le et al. [85] presented a measurement system with several wireless sensors to measure different environmental parameters in the greenhouse. Such sensors monitor the humidity of air, air temperature, soil moisture and the intensity of sunlight. Information that is gathered using such sensors is sent through wired networks to a Raspberry Pi, which acts as the main coordinator of analysis and local storage in an external microSD card. The Raspberry Pi is connected to a router via WiFi. The user interface is a specific site, with the help of which the user can see and work with the measurement data. The system was also tested for 1 week and proved that the results could be accurate when it came to monitoring various agricultural parameters within the greenhouse. Nonetheless, the wired connections are expensive, and this may be a disadvantage of scalability.

Pham et al. [86] introduced a mesh Wi-Fi wireless monitoring system platform that uses affordable hardware systems, such as the ESP8266 Platform. The sensors that were used at the six deployed nodes were used to measure the environmental temperature, soil moisture, soil temperature, light intensity and environmental humidity of the farm. The measured data were sent from nodes to the GW, which relays them to a server accessed by a website application running on a MongoDB, Express JS, AngularJS and Node.js stack, allowing the owner to track and save the information in real time. Communication in this system used the Wi-Fi protocol, and three station repeaters maintained connectivity. The current consumption between two deployed nodes in line of sight (LOS) over 15m was 74 mA, with each node powered by a small 3.7 V lithium polymer battery. The system was tested on the Vietnamese German University campus—an ideal greenhouse. The battery lasted just 55 h or about 2.5 days. This work can serve as a basis for future studies on IoT systems to help farmers monitor the parameters required for their farms. However, this system was limited to tracking greenhouses.

Several studies have highlighted many applications of WSNs in agriculture to reduce power consumption. Haseeb et al. [30] presented energy-efficient IoT-based WSNs for innovative agricultural applications. The system includes several agricultural sensors that aim to capture data and select cluster heads as part of a reliability-based multicriteria decision-making task. The signal-to-noise ratio was used to evaluate the data transmission efficiency based on signal strength. In addition, the security and throughput of data transmission between sensors and the base station are guaranteed by the linear congruential generator's period. The system was simulated on an agricultural field size of 200×200 m^2 using 100 sensor nodes. The results revealed that the system's communication performance improved across throughput, packet drop ratio, network latency, energy consumption and routing overhead, achieving 13.5%, 38.5%, 13.5%, 16% and 26%, respectively. However, the system was limited to simulation, and the type of wireless communication protocol was not determined.

The survey above reveals that various wireless communication systems possess distinct strengths and weaknesses suited for agricultural applications as shown in Table 2. LoRa/LoRaWAN technology offers a range of 2–20 km and consumes 66 mW/h of power, making it ideal for large fields. However, it faces challenges with data processing speed and model testing. Sigfox provides basic data transfer, flexible power use and extensive coverage of 10–40 km, but its limited data capacity and transaction costs reduce its practicality. NB-IoT features high power consumption, higher costs and high latency, enabling rapid data transfer with minimal delays, though it operates at elevated power levels. BLE technology stands out for its low cost, low latency and power consumption. Nevertheless, it operates within a limited range of 100m, making it suitable mainly for small operations or greenhouses. ZigBee's networking capabilities include low power consumption and mesh networking, but its signal range is limited by interference in the 2.4 GHz band. Z-Wave's sub-GHz frequency helps reduce interference, but this technology suffers from limited range and a single vulnerable point. RFID is an effective tracking solution for monitoring animals, though it requires minimal power and has range and data-throughput limitations. Large agricultural fields do not benefit from WiFi because it offers high data speeds and scalability at the expense of excessive power consumption and limited range. Technology selection is driven by specific requirements, including operational range, data transmission rates, power consumption and pricing, because no single solution suffices for agricultural use cases.

5.2 | Comparison of Previous Works-Based WSNs in Agriculture

The previous section introduced WSNs, demonstrating literature surveys in the last 5 years, which cover their applications in the agriculture sector involving monitoring, communication, management and improving production yields via smart decisions of intelligent farming with many different solutions to reduce the power consumption, which is the main challenge faced by WSN applications. Table 3 compares research from previous literature on measurement parameters such as temperature, humidity and soil moisture. It also discusses which sensors were used and their power-saving and power-consumption characteristics. The comparison was based on wireless communication protocol, processor type, sensor type, power type, methodology, performance metrics, field size and limitations. Based on recent studies, the essential requirements introduced in the previous surveys offer an overview of the technologies that have reduced power consumption and the limitations identified for future study.

As a recommendation, many researchers have proposed a system framework to reduce power consumption [30]. At the same time, some researchers used a low-power farm environment, as in refs. [35, 37], whereas others used low-power components, such as LoRa technology, as in refs. [36, 41, 65]. Solar panels and other technologies play an essential role in reducing power consumption as reported in refs. [63, 65, 66]. Furthermore, different methods were used to reduce power consumption, such as ref. [42], which reduced device usage; others improved traditional algorithms and others proposed new ones, such as refs. [64, 66]. Accordingly, LoRa is used for large-scale applications and ZigBee for small-scale applications as they offer similar coverage and low power consumption; however, both have low throughput. Previous research was often limited to simulation, such as in refs. [30, 37, 64]. Most did not consider the field's size, such as in refs. [35, 37, 64, 67]. On the other hand, Arduino was primarily used as a microcontroller.

5.3 | Path Loss Models of WSNs in Agriculture

Wireless agriculture communications, often called PA or smart agriculture, rely on sensors and wireless networks to monitor and manage agricultural activities. The effectiveness of these systems depends mainly on understanding the path-loss model, which describes how signals weaken as they propagate through the environment. Several considerations in this section have been reviewed in previous research and directly affect the signal path; they are as follows [87].

1. Distance: Signal strength decreases with increasing distance from the transmitter to the receiver.

TABLE 2 | WSN protocols comparison.

Ref./year	Technology	Frequency band	Communication range	Data rate	Power consumption	Network type	Latency	Strength in agriculture	Limitations in agriculture
Codeluppi et al. [36]/2020	LoRa/LoRaWAN	868/915 MHz (sub-GHz ISM)	2–20 km (rural)	0.3–50 kbps	Very low (≈66 mW/h)	LPWAN	33.67 ms	• Long-range, low power and cost-effective for vast fields • Suitable for remote crop, soil and weather monitoring	• Low data rate • Limited capacity for real-time or high-data applications
Ghori et al. [51]/2020	Bluetooth (BLE)	2.4 GHz ISM	10–100 m	Up to 2 Mbps	Very low	PAN (mesh)	6 ms	• Low power and cost • Easy smartphone integration • Ideal for greenhouse monitoring	• Very short range • Limited scalability for large farms
Tsakalidi et al. [48]/2021	NB-IoT	Licenced LTE bands (700–2100 MHz)	1–10 km	Up to 250 kbps	Moderate to high	Cellular LPWAN	1.6–10 s	• Reliable connectivity, low latency • Supports real-time PA and automation	• Higher energy consumption • Requires operator subscription • Limited in areas without LTE coverage
Rennane et al. [78]/2021	RFID	LF (125 kHz), HF (13.56 MHz), UHF (860–960 MHz)	Passive: < 5 m Active: Up to 100 m	Very low	Passive, none, active, moderate	Point-to-point	36 ms	• Ideal for livestock tracking, equipment ID • Maintenance-free for passive tags	• Very short range • No real-time data • Limited sensing capability
Le et al. [85]/2021	WiFi	2.4/5 GHz	20–100 m	Up to 600 Mbps	High	WLAN	50 ms	• High data rate • Suitable for image/video-based monitoring	• High power use • Limited range • Not ideal for battery-powered sensors
Haque et al. [59]/2022	ZigBee	2.4 GHz ISM	10–100 m	20–250 kbps	Low	WPAN (mesh)	20–30 ms	• Low-cost, mesh network extends coverage • Reliable for greenhouse automation	• Interference at 2.4 GHz • Short range and low data throughput

(Continues)

TABLE 2 | (Continued)

Ref./year	Technology	Frequency band	Communication range	Data rate	Power consumption	Network type	Latency	Strength in agriculture	Limitations in agriculture
Yamazaki et al. [43]/2023	Sigfox	868/902 MHz	10–40 km (rural)	100 bps (uplink)	Ultra-low	LPWAN	2.5–4.5 s	• Excellent coverage, low cost and power • Ideal for low-data sensors (moisture and temperature)	• Very small payload (≤ 12 bytes) • Limited number of messages/day • Not suitable for control systems
Effah et al. [72]/2023	Z-wave	868/908 MHz	30–100 m	Up to 100 kbps	Low	Mesh	35–60 ms	• Operates on sub-GHz → less interference • Stable network for indoor farming	• Short range • Fewer nodes per network • Not suited for large-scale outdoor fields

2. Obstacles: Physical obstructions, such as trees, crops, buildings and terrain features, can cause signal reflection, diffraction and scattering, leading to additional path loss.
3. Weather conditions: Rain, humidity and other factors can affect signal strength, particularly at higher frequencies.
4. Antenna height and type: The height and type of antennas used in the communication system can influence path loss. Higher antennas typically have less obstruction and, therefore, lower path loss.
5. Frequency: Higher-frequency signals generally experience greater path loss than lower-frequency signals.
6. Vegetation density: Crop type and density can significantly affect signal propagation. Dense vegetation tends to absorb and scatter signals more than sparse vegetation.

Therefore, PLMs are essential in designing and implementing WSNs in agricultural applications. These models help estimate reductions in signal strength caused by various environmental factors. Below are the common types of path loss models relevant to the agricultural sector.

5.3.1 | Free Space Path Loss Model (FS-PLM)

FS-PLM is a path loss prediction model for a wave outward as a spherical wave without obstacles in free space, such as a vacuum or ideal open space. It is used for short-range communication in environments with minimal interference, such as open fields during early crop growth stages with little vegetation. However, FS-PLM is unrealistic for most agricultural scenarios due to obstacles such as crops, trees or terrain undulations. The equation related to FS-PLM is expressed in Equation (1) [88].

$$\mathrm{PL}_{\mathrm{FS}}(\mathrm{dB}) = 32.44 + 20\log F + 20\log D \tag{1}$$

where D is the distance between transmitter and receiver (in metres) and F is the frequency (in MHz).

Anzum et al. [88] presented a path-loss model based on empirical received signal strength measurements from a palm oil plant to model the LoRa channel. The experiment was conducted on farmland with an area of (2×0.5) km^2, where data were collected using LoRa ESP32 devices (used as transmitters and receivers) for propagation measurements of 1500 line-of-sight and 300 nonline-of-sight paths to develop a path-loss prediction model. In the line-of-sight measurement, the transmitter and receiver were positioned close to the ground. In contrast, the transmitter and receiver were elevated to 1.5 m above the ground for the nonline-of-sight measurement to assess signal propagation through the canopy and trunk due to the plantation of the adult trees comprised of multilines of trees, which in role acts as an intervening wall for the propagation path in nonline-of-sight, the proposes of the PLM for palm-oil plantations based on Free space model with the addition of summation of the attenuations caused by number of canopies and trunks. The model was tested at two bandwidths, 125–250 and 500 kHz, resulting in 2.34 and 2.9, respectively, for the line-of-sight predicted propagation model. In contrast, at

TABLE 3 | Comparison of previous studies based on WSN in agriculture.

Ref./year	Wireless technology	Controller type	Sensor type	Powered type	Methodology based	Performance metrics		Size of the field	Limitation
Sharma et al. [63]/2019	ZigBee and WiFi	Arduino UNO board	Temperature and humidity	Alkaline 1.5 V with ambient solar energy	Solar energy harvesting technique	Power consumption	20.62%	20 × 50 m^2	Limited to simulation
						Network lifetime	20.13%		
						Throughput	1.6%		
Ahumada et al. [41]/2019	Sigfox and LoRa	ESP32	SHT 15	Li-ion battery with photovoltaic cell	Low power components	Provide the sensor node lifetime independently	5 years	37.66 ha	Limited automation in rural areas
Haseeb et al. [30]/2020	Heterogeneous	N/A.	Temperature, humidity water and moisture levels	N/A	Proposed an IoT-based WSN framework comprising different design levels.	Network throughput	13.5%	200 × 200 m^2	Limited to simulation
						Energy consumption	16%		
						Packet drop ratio	38.5%		
						Network latency	13.5%		
						Routing overheads	26%		
Yang et al. [35]/2020	LoRa	STM32F103ZE microcontroller	CCS881, GY302, HDC1080 and YL69	N/A	Low-power farm environment detection system based on LoRa	Working current in sleep mode	0.3 μA	N/A	Limited to the greenhouse
Codeluppi et al. [36]/2020	LoRaWAN	LoPy4 board and CC3200 and Raspberry	DS18B20, and AM2032, respectively, in addition to SHT-10	3.7 V LiPo, 1800 mA with solar LiPo charger	Low power components	Reduce power consumption	From 14 mA/h to 9 mA/h	Greenhouse of 20 × 9 × 5 m^3	Unreliable measurements
PITU et al. [42]/2020	Sigfox	EspressifES32	SI7006-A20, MPL3115A2 and LTR 329ALS-01	N/A	Reducing the resources used	Reduce current consumption	Up to 15 mA	N/A	The system was costly and unsecured
Agrawal et al. [64]/2020	ZigBee, LoRaWAN	N/A	Temperature, humidity, soil, wind direction and wind speed sensors	Solar panel	Propose a product density model and improved duty cycling	Power consumption	6%–7%	N/A	Limited to simulation

(Continues)

TABLE 3 | (Continued)

Ref./year	Wireless technology	Controller type	Sensor type	Powered type	Methodology based	Performance metrics		Size of the field	Limitation
Sadowskia et al. [65]/ 2020	LoRaWAN, ZigBee and WiFi.	Arduino UNO	Grove soil moisture sensor	Grand-Pro 3.7 V 6600 mAh Lithium Polymer (LiPo) with star solar D165X165 monocrystalline solar panel	Long range wireless area network with energy harvesting	Average current consumption	69.36, 29.33 and 171.17 mA	1500, 120 and 50 m	Require extra hardware and high power consumption
Lozoya et al. [66]/ 2021	ZigBee	Arduino and Raspberry Pi 3	10HS, PYR sensor, Davis Cup sensor and VP4 sensor	Solar radiation sensor	Self-triggered algorithm	Power consumption	20%	25×100 m^2	This system had high irrigation errors.
						Communication messages	85%		
Morais et al. [67]/ 2021	IEEE 802.15.4/ ZigBee, LoRa/ LoRaWAN and GRPS	RISC 32-bit	SPWAS'21 devices	Li-Po battery with solar panel	Low-cost, flexible and robust data acquisition device	The average current in active mode	140 µA	234×160 m^2	Data exchange, together with further miniaturisation is required
						The average current in sleep mode	83 µA		
						Cost	100 EUR		
Prasad et al. [37]/ 2022	LoRa	Arduino	Temperature, humidity and soil moisture	N/A	Low-power farm environment detection system	Reduce power consumption	66 m W/h	N/A	Limited to simulation
López et al. [38]/2023	LoRa	MSP430FR5994 MCU	SHT10 and SEN0169	BQ25570	Develop a sustainable forage grass power fuel cell	Power consumption	1.51 mW	N/A	Limited power generation
						Power conservation	19%		
Yamazaki et al. [43]/ 2023	Sigfox	Arduino (UNO R3)	Temperature and humidity, illuminance and CO_2	Two AA-N-MH	Improving embedded software and leakage current countermeasures of electronic circuits	Extend battery life time	5.7 years	26.35 km	The required current consumption was unstable among sensor nodes

nonline-of-sight propagation, the attenuations for the available spreading factors of 7 and 12 were found as 7.58 dB, 7.04 dB, 5.35 dB, 5.02 dB, 5.01 dB and 5 dB for trunks and 9.32 dB, 7.96 dB, 6.2 dB, 5.89 dB, 5.79 dB and 5.45 dB for canopies. In addition, the proposed model was evaluated for predicting PLM in palm oil plantations by comparing it with Weissberger's and ITU-R, which serve as empirical foliage-loss models, achieving a lower mean RMSE of 2.74 dB. Thus, it is considered the best choice. However, the model is specified for a limited frequency range.

5.3.2 | COST-235 Model

The COST-235 model is the most popular foliage path-loss prediction model, based on exponential-decay model simulations in MATLAB. It estimates signal attenuation due to vegetation based on canopy depth. It applies it to mobile communication systems operating within 1–10 GHz of the work frequency, including Earth-space and land mobile systems. This model is an ideal choice because of its accuracy and ease of adaptation to different conditions, such as cloud, rain and gaseous absorption, which can affect attenuated rates. The COST-235 PLM is expressed in Equation (2) [89].

$$PL_{\text{COST-235}}(\text{dB}) = \begin{cases} 26.6 \times F^{0.2} \times D^{0.5} & \text{out of the leaves} \\ 15.6 \times F^{-0.009} \times D^{0.26} & \text{on the leaves} \end{cases} \quad (2)$$

where D is the distance between the transmitting and receiving antennas in metres and F is the operation frequency in MHz.

Dogan et al. [89] presented a novelty of empirical PLM at 5G frequencies of (3.5 and 4.2 GHz) with foliage depth based on the tree's volumetric-density rate in the citrus orchards for WSN. The empirical PLM was proposed to determine the impact of volumetric density rate, in addition to distance and frequency, for a 30,000 m^2 orange orchard, with measurements implemented across multiple regions with the same environmental parameters to improve accuracy. At first, the path loss was generated from initial measurements across different environments and compared with LITU-R, FITU-R, Cost 235 and log-normal models. The generated path loss was measured at 3.5 and 4.2 GHz, yielding R^2 values of 0.966 and 0.965, respectively. Concurrently, the generated PLM was developed depending on volumetric density rate, distance and frequency. The proposed model was tested, showing a 20 dB reduction in path loss alongside a 30% increase in volumetric density. The volumetric density rate was estimated to be 0.571 based on the four empirical models referenced. Additionally, the RMSE for the generated and suggested PLMs were 3.08 and 3.46, respectively. However, it was limited to specific frequencies.

5.3.3 | Two-Ray Ground Reflection Model

It is described as a practical PLM since it accounts for both directed and reflected signals from the ground. It plays an essential role in agriculture because it performs well on flat terrain with a clear LOS, such as paddy fields or plains, and is ideal for low-vegetation areas where ground reflection is prominent. However, this model is inaccurate for hilly terrains or dense foliage as it does not account for signal diffraction or scattering. The two-ray ground reflection-PLM is described in Equation (3) [90].

$$PL_{\text{T-RGR}}(\text{dB}) = 40 \log_{10}(D) - 20 \log_{10}(H_t) - 20 \log_{10}(H_r) \quad (3)$$

where D is the distance in metres that separates the receiver and transmitter antennas, and H_t and H_r are the heights of the transmitter and the receiver antenna in metres.

Adi et al. [90] analysed the transmission data for ZigBee 2.4 GHz and LoRa at 915 and 920 MHz in snow conditions. The data were measured using a two-ray ground reflection model that compared the propagation of ZigBee and LoRa. The transmitter and receiver were placed in a mountainous area covered in snow to obtain signal reflections. This approach's analysis concluded that the heights of the sender and receiver affect the two-ray ground reflection model. In addition, comparison results indicated that the signal strength is the same for LoRa and ZigBee. However, ZigBee was more efficient at a distance of less than 120 m.

5.3.4 | ITU-R Model

The ITU-R model is another path-loss propagation model based on the exponential decay model, used to estimate signal attenuation due to vegetation based on canopy depth. Similar to the COST-235 model, it is ideal for dense crop fields, forests or orchards with vegetation. In addition, this model is flexible in rural and semirural geography; however, its work is limited to site-specific vegetation data, such as density and leaf moisture content. The ITU-R PLM is outlined in Equations (4) and (5) [91].

$$PL_{\text{ITU-R}}(\text{dB}) = R_{\text{foliage}}(\text{dB}) + PL_{\text{T-RGR}}(\text{dB}) \quad (4)$$

$$PL_{\text{ITU-R}} = 0.2 \times F^{0.3} \times D^{0.6} + 40 \log D - 20 \log H_t - 20 \log H_r \quad (5)$$

where R_{foliage} is equal to $(0.2 \times F^{0.3} \times D^{0.6})$ in dB, D is the distance measured in metres between transmitter and receiver and F is the operating frequency in MHz.

Gonsioroski et al. [91] evaluated the measurement campaign of two urban and four vegetation PLMs in an urban environment, including living areas with extensive vegetation, squares and urban parks. They are associated with IoT technologies in the future. Using the IEEE 802.11ah standard in the 900 MHz band in such an environment enables modelling of both short- and long-term channel fading for IEEE 802.11ah system planning and prediction, which are the objectives of IoT applications in this environment. The experiment was conducted in a living environment with multiple bushes and large trees, with varying foliage density, cars, people, tables and parking, to collect empirical data. Omnidirectional antennas with 3 dBi were placed at both the transmitter and receiver, at 7 and 1.5 m, respectively. The outdoor PLMs for the IEEE 802.11 task group include two scenarios: macro deployment and pico/Hot-zone deployment. The acquired data were analysed to determine the optimal path-loss model for small-scale and large-scale fading statistics. The PLMs were based on comparing SUI and the COST 231-Hata models used in urban environments. On the other hand, vegetation PLMs from Weissberger's model, COST235 and the Fitted ITU-R models were selected for comparison. An ITU-R PLM modification was applied, reducing the RMSE by 6.18, the lowest mean error among the other models.

5.3.5 | Log-Distance Path Loss Model

A generalised PLM introduces a path loss exponent for different environments, adjusting for specific agricultural environments by calibrating it to (2–2.5) for open fields and (3–4) for dense crops. It can model urban, suburban and rural environments. It is versatile for estimating path loss in mixed environments, such as fields with uneven crop density or moderate vegetation, and estimating communication range for sensor nodes within agricultural IoT systems. However, calibration of the path-loss exponent using empirical data from specific fields is required, and it cannot mitigate mobile shadowing effects caused by animals and other farm equipment. The log-distance PLM is presented in Equation (6) [92].

$$PL_{\text{L-DPL}}(\text{dB}) = PL(D_0) + 10\beta \log_{10}(D/D_0) \tag{6}$$

where PL (D_0) is the path loss over reference distance D_0 (1 m), β is a path loss exponent signifying how rapidly the path loss increases with distance and D is the distance between the transmitter and receiver in metres.

Phaiboon et al. [92] developed two empirical PLMs operating at 433 MHz to plan LPWAN for farmland with Ruby mango vegetation and to determine path loss at the canopy and trunk levels. There were 320 trees of Ruby mango planted in straight lines between each line 6m, where each trunk in the same line separated by 5m, where the LoRa was used as transmitter and receiver to collect data in line-of-sight and nonline-of-sight (trunk and canopy levels which acts as attenuation factor) with spread factor of 7 and bandwidth of 125 kHz to attain the best receive signal strength indicator. Additionally, omnidirectional antennas were used at each transmitter and receiver height to assess the impact of the surrounding environment on the propagation model, which was placed at heights of 0.3, 1.2, 2.2 and 2.7 m above the ground. The free-space model was used as a reference to inform both proposed models. Based on the ITU-R model and the two-ray ground reflection model, the free-space model serves as a reference for evaluating path loss in various environments. In addition, another PLM, named tree attenuation factors based on log distance, was proposed.

The work was compared for evaluation with other empirical PLMs from ITU, COST and FITU, which resulted in MAE values of (3.84, 4.58) and RMSE values of (4.93, 5.9) for both models. Results achieved higher accuracy than ABC-PLM, making it more suitable for typical Ruby mango tree patterns. However, there were prediction errors for flowers and fruits on the trees.

5.3.6 | Log-Normal Shadowing Path Loss Model (LNSM-PLM)

This PLM expands the log-distance model by a Gaussian random variable for shadowing from obstacles, which impact the signal strength in fluctuations. It is used in orchards or irregularly planted fields to capture variability caused by random barriers such as trees, plants, terrain irregularities and environmental heterogeneity. This model suits environments with irregular crop distributions, orchards or small-scale greenhouses, and it can predict communication performance under varying vegetation densities. However, it tends to increase complexity due to the stochastic component. The LNSM-PLM is expressed in [93] as illustrated in Equation (7).

$$PL(D)\,(\text{dB}) = PL(D_0) + 10\beta \log_{10}(D/D_0) + X_\sigma \tag{7}$$

where the PL(D) denotes the path loss, X_σ is the Gaussian random variable with zero mean and σ is the standard deviation. The samples β and σ are called the path loss at a reference distance D_0.

Pal et al. [93] developed a PLM by analysing the strength of the communication signal between two sensor nodes and how it is affected by plant height and density under standard IEEE 802.15.4 wireless communication. The path-loss coefficients of millet and rice crops have been measured and analysed for 2.4 GHz RF signals to investigate the distribution of sensor nodes and the effect of sensor separation on network connectivity. Additionally, to improve RF modelling accuracy, the experiment was conducted in two fields using a WSN framework in two scenarios: the first with all nodes positioned 52 cm above the ground, a real-time operating system and a radio duty cycle protocol for power consumption management. In contrast, in the second scenario, the sensor nodes were placed above the ground at 52 cm. The first scenario, a monitoring-based Voronoi diagram for node deployment strategy, summarises the first and second experiments and tree-based PLM, which collect 1200 received signal strength samples over 126 days. At the same time, the second scenario analysed the impact of a furrow on the crop target, collecting 1200 received signal strength samples over 640 days. The measurement of Received signal strength in rice and millet fields was conducted over areas of 7.53×10^4 m^2 and 8.04×10^4 m^2, respectively.

Additionally, a linear regression approach was used to estimate the path loss exponent, thereby reducing the error between the empirical and estimated received signal strength. Accordingly, the bit error rate for the distance between the transmitter point and the transmission range $< 2\%$ was 58 m, with a minimum signal level in free space of -75 dB. Also, the disconnectivity in both experiments occurred at a maximum separation distance of 58m and an average plant elevation of $\approx$58 cm. The results were due to the path loss exponent equal to 1.63, which meets the separation distance of the free-space transmission signal level. After comparing tree-based PLM, it can be concluded that the bit error rate was more than 2% in vegetation depth of ≥ 15 m after the panicle stage and after the flag-leaf stage of rice crop and millet crop, respectively. However, additional factors affecting the transmission signals between sensor nodes are required.

5.3.7 | Okumura–Hata Model

Okumura–Hata is an empirical model for predicting signal propagation in large-scale urban, suburban and rural areas, on flat terrain or in areas with sparse vegetation. It can be adjusted for different antenna heights and environments. This model is widely used in large rural fields or plantations for long-range communication with a transmission range of 20 km. In addition, it is suitable for deploying PA base stations. Furthermore, the prediction accuracy of this model depends on carrier frequency, distance and terrain corrections that account for rural conditions, transmitter and/or receiver heights, mobile antenna height and base station antenna height [94]. The Okumura–Hata model can be used to evaluate the efficiency of communication networks for wireless sensors, UAVs and IoT devices used in precision farming. Due to its efficiency and simplicity, it can be used to plan networks in sparsely populated areas, including agricultural areas. However, it is inaccurate at high frequencies (> 2 GHz) or in environments with high vegetation density as it is limited to 150–1500 MHz. The mathematical expression of the Okumura–Hata model is introduced in Equations (8) and (9) [95].

$$\begin{aligned} \mathrm{PL}_{\text{Okumura-Hata}} = 69.5 + 26.16 \log_{10} F - 13.82 \log_{10} h_t - \mathrm{CF} \\ + \left[44.9 - 6.55 \log_{10} h_t\right] \log_{10} D \end{aligned} \tag{8}$$

$$\mathrm{CF} = 0.8 + \left(1.1 \log_{10} F - 0.7\right) h_r - 1.56 \log_{10} F \tag{9}$$

where $\mathrm{PL}_{\text{Okumura-Hata}}$ is the Okumura–Hata PLM, which is measured in dB in rural areas, and D is the distance in kilometres between the receiver and transmitter. In addition, the antenna heights are in metres and represented by h_t for the transmitter and hr for the receiver; CF is an antenna parameter with a height correction factor given by Equation (9).

Shakir et al. [95] compared path-loss measurements for LTE data in urban environments using various propagation models. A comparison of multiple models—free-space path loss, two-ray, log-distance, Lee, Okumura and Okumura–Hata—was applied to evaluate the best-performing model for this urban environment. The LTE-measured data were collected at regular intervals of 100 m using RF drive tests in the 700 MHz frequency band, with a mobile station height of 1.5 m and a gain of 0 dB. The measured data were evaluated using free-space path loss, Okumura, Okumura–Hata, two-ray and Lee models and log-distance models to determine which is best for prediction in urban environments. The results of RMSE and MAPE were (46.6 dB and 35%), (75.39 dB and 58%), (32 dB and 24%), (22.3 dB and 17%) and (16.3 dB and 12% w) for free space, two-ray, Okumura, Lee and log-distance models, respectively. However, the Okumura–Hata model achieved the best accuracy, with 4.23 dB and 3% RMSE and MAPE, respectively.

5.3.8 | GUT-R Model

Unlike other PLMs, the GUT-R is used to model propagation losses for UAV data acquisition and aerial surveillance, accounting for two-ray ground-reflection effects, which are essential in open agricultural fields. This PLM-based GS height, antenna gain, UAV height, ground reflected, direct line-of-sight and horizontal separation distance. It is suitable for path loss estimation in rural and semirural environments. The GS-UAV model is illustrated in Equation (10) [96].

$$\begin{aligned} \mathrm{PL}_{\text{GUT-R}}(\mathrm{dB}) = 40 \log_{10} D - 10 \left(\log_{10} G_{gs} + \log_{10} G_{\mathrm{UAV}}\right) \\ + 20\left(\log_{10} h_g + \log_{10} h_{\mathrm{UAV}}\right) \end{aligned} \tag{10}$$

where G_{gs} and G_{UAV} represent the antenna gain of GS and UAV, respectively. The h_{UAV} and h_g denote the UAV altitude and GS height, respectively.

Duangsuwan et al. [96] proposed a PLM based on machine learning (ML) to characterise two-way communication between ground sensors and UAVs in a smart farm, named the GUT-R model. Napier and Ruzi grass farms were measured using support vector regression (SVR) and artificial neural network (ANN) models, and the results were compared with the empirical GUT-R model. It resulted in R^2 values ranging from 0.973 to 0.983 for ANN and from 0.920 to 0.954 for SVR. ANN performed better than SVR in all cases, achieving an accuracy of 97%. In addition, the effectiveness of the GUT-R model varied by farm type and UAV altitude. However, the GUT-R model requires large-scale data and is sensitive to environmental conditions.

Among the listed path loss models, the LNSM-PLM is the best option, as it quantifies attenuation based on distance and unpredictable shadowing from obstacles such as trees and plants. The LNSM-PLM addresses signal variations in actual agricultural and forestry environments, where signal intensities are highly variable. Including a random Gaussian variable makes this prediction method generate more realistic signal behaviour, thereby minimising unanticipated data transmission losses. Data transmission loss reaches its maximum in practice because the FS-PLM relies on the theoretical assumption of obstacle-free conditions, which do not match real-world scenarios. FS-PLM should be avoided in deployments in agricultural and rural areas because it fails to account for natural elements, such as vegetation, physical features and interference, which are common sources of problems. An advantage of FS-PLM is its theoretical ease of use. However, the model fails to deliver the necessary robustness for dependable data transmission in operational environments.

6 | Previous Works-Based Drones in Agriculture

UAVs are among the most important solutions for capacity improvement due to their cost-effectiveness, pop-up fashion and ease of implementation, which make them a mobile base station [97]. UAVs are classified into two categories: fixed-wing drones and rotary-wing drones, each selected based on the environment, the required quality of service and the federal regulations required for that quality of service [98]. Fixed-wing UAVs, such as small aircraft, have high speed and weight and require moving in one direction to stay aloft. In contrast, rotary-wing UAVs, such as quadrotor drones, can lift off, propel forward and backwards and hover over a given location [99]. However, quadcopters improve performance in large agricultural fields by monitoring crops and spraying pesticides, thereby minimising pesticide use and fertiliser inputs and increasing crop yield [16].

Specifically in the agriculture sector, UAVs are characterised by precision and real-time data, which is increasing their daily popularity. The best applications and practices of UAVs that support smart agriculture include estimating soil conditions, planting future crops, controlling pests and diseases, crop surveillance, livestock monitoring and agricultural spraying [100].

It is worth mentioning that there is a trade-off between payload and maximum flight time. So, if the latter increases (e.g., through reinforcement with powerful batteries or a large fuel tank), the weight will increase, thereby decreasing the payload [101]. Accordingly, minimising the weight of measurement instruments and reducing the need for additional equipment has enabled the development of UAV sensor systems and expanded their application fields. The primary sensor classifications are clustered in several sets.

1. Sensors that are mainly used for monitoring and localisation include multispectral sensors, red, green and blue (RGB) sensors (visible spectrum), hyperspectral sensors, optical sensors, thermal infrared sensors, radar sensors, consumer-grade sensors, light detection and ranging (LiDAR) sensors, inertial measurement unit, GPS and magnetometer [102].
2. Meteorological sensors are another type of sensor that can be added to measure environmental parameters, such as humidity, temperature, gas emissions and crop conditions [102].
3. Biological sensors, chemical sensors, spraying systems or similar payloads are required in some smart agriculture applications involving crop treatment, such as fertiliser and irrigation, underscoring the need to equip UAVs with tanks and end-effectors as nozzles [103].

Since the sensor nodes are constrained in dimension and cost, some agriculture applications based on a new topology of flat ad-hoc networks are introduced, which represent multiple UAVs forming a group with self-initiating communication related to a satellite or ground base station that can be used specifically for crops suffering from harmful microscopic organisms by treating them with urea and various pesticides [104]. Different types of UAV technology are used across different agricultural applications, with and without IoT agriculture systems, as shown in the survey below.

6.1 | Matrices Series

Zhang et al. [105] presented a novel IoT system for remotely monitoring the environment using a UAV, LoRa technology and 5 GHz communication to enable high-speed data collection. The UAV-based board relay collects environmental data from IoT monitoring devices deployed in harsh areas via 5 GHz wireless technology, improving data collection speed in such regions and validating the LoRa performance module's ability to wake the 5 GHz module and monitor the region of interest remotely. On the intelligent groundside, which includes several sensors, a data logger, a control unit, a 5 GHz module and a long-distance wireless communication LoRa module, the sensed data are stored in SD-RAM. When the drone enters the target region, it sends instructions via the UAV's LoRa module to the ground LoRa to wake up the 5 GHz communication. When the drone hovers over the recommended region in the trajectory plan, the data are sent to the UAV via LoRa on the ground. Then, the UAV will fly to the next point to collect data, offload stored data or reach another region with a public network to forward it, or save it to a data centre on the ground. The system successfully collected data with a throughput of 3.5 Mb/s at a height range of 140m. However, the system has a limited capacity of the upper communication link, which will increase the transmission time, and the received signal strength will decrease with increasing distance. Additionally, the author did not mention the drone's battery capacity or flight time, which are among the main challenges in UAV applications.

Awais et al. [106] presented a platform for practical UAV experiences to support remote PA mapping and sensing. The demand for UAVs was to determine the optimal time and altitude for thermography using high-resolution remote sensing over the tea plantation. To collect images, UAV was compact with RGB and thermal camera, where the thermal camera was characterised by a resolution of (640 × 512 pixels), sensitivity of (50 MK @ f/1.0), field of view of (57.120 × 42.440), detector pitch of sensor was 17 μm and thermometric range (−40°C–150°C) and (−40°C–550°C) for high gain mode and low gain mode, respectively. The flight tests ranged from 2 to 5 m/s at 09:00, 11:00 and 13:00 h, reaching heights of 25 m, 40 m and 60 m, respectively. That would effectively calibrate the leaves and canopy with precision using an infrared thermometer. Real-time kinematic technology collected accurate GPS coordinates, which served as ground control points. The field was divided into a point grid using control points distributed along the edge and within the field at 50 m intervals. The orthomosaic received its geometric correction during image processing by incorporating those points. A process of orthomosaic generation has been applied, in which the original imagery pixels are precisely matched to each actual interval, colour and near-infrared orthophotograph. The proposed experimental platform showed that the most suitable altitude for taking an infrared image is 60 m at 11:00 w, with the strongest relationship, and that, to identify plant water stress, the best height and time to capture an image are 60 m and 13:00, respectively. Finally, the best time and height for canopy temperature determination are 11:00 and 60 m, respectively. However, there was a limitation on UAV flight time.

Xu et al. [107] presented a design for an open-source UAV equipped with a LiDAR sensor and three types of cameras—thermal, RGB and multispectral—for precision crop management and plant phenotyping. UAV is adopted to build a flexible system for data acquisition, transferable data and processing data in the pipelining method to collect data from a cotton field divided into 96 sections (8 columns and 12 rows) where there is an alley with a length of 1.5 m between the plot and also a distance of 1.8 m between crops, UAV equipped with data acquisition system (second version) and flight according to planning path with height of 25 m, speed of 1 m/s and 70%, 90% for side overlap and forward overlap, respectively. Ground data were collected with a portable normalised difference vegetation index sensor, and thermal calibration was used to measure temperature and serve as a ground control point. Data preprocessing, image calibration, georeferenced orthorectified mosaics and 3D model creation can be partially performed on the UAV's onboard computer in real time. Morphological traits, vegetation indices and canopy temperature were extracted to assess several phenotypic traits. Furthermore, the correction was implemented using Meta Shape and Python. The results of such a system are 1.02°C, 6.6% and 0.1 m for temperatures derived from thermal images, the mean relative error of the canopy normalised difference vegetation index and the maximum height error of the canopy, respectively. However, such a design has limited flight time.

Awais et al. [108] highlighted the integration of very high-resolution RGB and thermal imagery collected by UAVs to improve the possibilities of canopy temperature detection. To provide a comparison between an accurate canopy temperature and thermography performance evaluation, a multirotor UAV compact with RGB thermal and cameras with a weighted battery of 10 kg, altitude of 60 m, speed of 7 ms^{-1} and space gap of 6.12 cm was used over 5-acre tea field to collect RGB and thermal images; at the same time, the temperature of shady and five sunny leaves was measured by handheld thermometer. Additionally, physiological parameters were measured on the sampling day, including ground-based canopy temperature, stomatal conductance, leaf area and soil water content. On the other hand, the 2-h sunset leaf area index was calculated. The image processing is described, and canopy temperature measurements from thermal images have been carried out in MATLAB. In contrast, the calibration of the thermal temperature has been executed by the Flair software. Accordingly, calibration of the extracted canopy temperature and the UAV's thermal sensors yielded a determination coefficient (R^2) of 0.9297. Moreover, the relationship between the crop water stress index and stomatal conductance has resulted in field capacities of 90%–100%, 75%, 60% and 50%. Canopy temperature increased from 09:00 to 11:00, and then decreased to 13:00, whereas the R^2 values for time were 0.90, 0.75 and 0.86, respectively. In addition, the comparison of the R^2 values for canopy temperature, stomatal conductance and standard deviation yielded 0.755, 0.67 and 0.695, respectively. Finally, the most reliable estimated temperature approach resulted in about 35°C to 40°C.

Cariou et al. [105] presented a UAV equipped with a data collector to collect data from buried sensor nodes. The objective for the UAV was to improve data collection from the underground sensor in a feasible manner, experimentally, via a LoRa communication module and retransmitting it via ZigBee (which was compacted with the sensor node on the UAV) to the ground gateway, which acted as a ZigBee-Pro interface with the network server via Ethernet. At a depth of 15 cm, the sensor nodes were buried to measure the soil parameters and store them in memory. Then, these data were transferred to the UAV by triggering the transmitting process with a push button connected to the underground sensors once hovered to collect data over the appropriate 21 positions of a proposed planning path of flight to reduce power consumption at a speed of 23 m/s for 40 min with the altitude of 20 m, 40 m and 60 m to evaluate the impact. The experimental results show that the received signal strength indicator is maximised when the UAV at the highest altitude is set to more than 50 m. It is worth noting that two trajectory-planning algorithms were configured: an ant colony and a genetic algorithm, which resulted in execution times of 2.4 and 24.2 s, respectively. However, there were limitations in transmitting data from underground sensors to UAVs (about 0.29–5 kbps) and in UAV battery lifetime.

6.2 | Quadcopter

Castellanos et al. [109] presented an NB-IoT-based UAV as a mobile gateway to collect underground data on soil parameters of potato crops. To reduce power consumption in transmission and prolong battery life, UAVs used simulation tools to collect data from underground sensors. The experimental potato field area was 20 ha. Sensor nodes are deployed from 500 to 5000 to evaluate the impact of the node's intensity between each one, 10 m, with different buried depths of 0 cm, 25 cm, 50 cm, and 75 cm. Climate data and soil parameters were collected from sensor nodes via NB-IoT every 1 h while the drone flew at 12 m/s twice daily, then sent to the core network. Furthermore, mixed PLM was proposed. The results of this proposal are that when the sensor node density increases and the power consumption is reduced as well as when the buried depth of the sensor increases, the flight height of the drone increases, which increases the power consumption of the drone and the optimal altitude for the drone to collect data from buried sensor nodes is between 60 and 80 m. Additionally, from WSN's perspective, the batteries for sensors above ground and underground can be replaced every 82 and 77 months, respectively. However, the system is limited to simulation and requires analysing and processing data to make decisions with Machine Language technology.

Eladl et al. [110] proposed a conceptual UAV-based crop management sensing system that includes a new multistage classification system. The system shows good classification of crop types, including rice species and weed detection, using authentic UAV image datasets. It performed better, achieving up to 100% accuracy on the weed-net dataset, 99.5% on rice seedling and 97.99% on rice varieties, thereby outperforming the current state-of-the-art techniques. However, the small sample size, limited crop variety range, lack of standardised UAV testing protocols and absence of decision-support software to guide camera and platform selection are limitations of this study. Therefore, the use of UAVs remains limited, which can be explained by their cost-prohibitive nature and limited access to this technology among small-scale agrarian farmers.

Aszkowski et al. [111] presented a cost-saving practical UAV-based framework for evaluating wildlife damage in cornfields that relies on RGB images and deep learning (DL) models, including U-Net and SegFormer. The approach was trained on a dataset with over 300 ha in western Poland, with an intersection-over-union of 0.88 between healthy and damaged crops. The supporting processing code and trained model are publicly available. However, the research has limitations, including inconsistent annotations, a small training dataset and the need to supervise the model by an expert, which reduces model accuracy and reliability. In addition, the use of NDVI data resulted in insignificant gains, and possible extensions to satellite imagery and transformer-based models have not been studied.

6.3 | Fixed Wings

Heinemann et al. [112] presented an optimised algorithm for retrieving land-surface temperature that combines thermal UAV imagery with multispectral optical imagery and is used in operational agricultural applications for mapping heterogeneity and different crop systems. A fixed-wing UAV with low speed, down to 4.5 ms^{-1} and a height of 77 m features a novel sensor platform integrated with a thermal camera and a multispectral sensor to capture long-wave infrared, visible and near-infrared in real time and to activate agricultural areas through semi-supervised thermal mapping. The algorithm steps yielded a mean absolute error of 0.5k during the evaluation of thermal measurements. However, the UAV's power supply had a limitation.

6.4 | DJI Mavic Series

By developing a nondestructive method, Yamagishi et al. [113] estimated the number of tillers at vegetative and reproductive stages under flooded conditions. A commercial-grade UAV was based on collecting RGB images by 5472 × 3648 resolution of the camera on board and over the 6 × 7 m^2 rice field at an altitude of 3 m from a 1.5 m with an elevation angle of 35°, which were processed by pipelining the tillers and emerging leaves in rice morphogenesis synchronously by using Python in Google Collaborator. Additionally, on one day, 20 seedlings were transplanted at a hill density of 21.2 m^{-2}, with each hill containing 3 to 5 seedlings, to evaluate the method's effectiveness. The workflow-estimated tiller number is significantly correlated with the actual tiller number within a 1.1 m × 1.1 m area, resulting in errors at the early reproductive stage with an average of 1.02 and a standard deviation of 1.09. On the other hand, the error of 60% of hills was $< \pm$ 3 tillers. In sum, this novel method improves the practical acceptability of high-throughput phenotyping field tasks.

Sun et al. [114] estimated the strong influence of the rice leaf area index by using colour indices driven from RGB images collected by UAV, by the image background and illumination variations. To separate the effects of elimination variations and RGB background, Retinex correction, image segmentation and analysis, changes in colour indices were presented, with and without these methods. RGB digital camera equipped with UAV to gather images of rice field comprised of 24 plots, each with an area of 30 m^2 (5 × 6 m^2). The data collected by the UAV were implemented over five stages (tillering, grouting, elongation, booting and heading) at a high spatial resolution, with an altitude of 10 m, a flight speed of 3 m/s and forward and side overlap of 80% and 70%, respectively. Rice leaf area index data were also collected using destructive sampling. The workflow resulted in the sum: (i) low changes in colour indices affected the first two stages and vice versa, the last three stages were affected; (ii) at the elongation stage, the correlation between colour indices and increased of 0.12 based Retinex correction, (iii) at tillering stage, the estimation accuracy of leaf area index mode was improved by using Retinex-corrected colour indices which permits R^2 of 0.57, (iv) in most colour indices Significant differences between rice and background were observed, (v) after image segmentation, and at the booting, heading and filling stages, there was a correlation increasing between colour indices and leaf area index and (vi) image segmentation has a most outstanding efficient accuracy over all stages. However, there was a limitation in UAV power supply and flight time.

6.5 | Other Types

Gao et al. [115] presented an IoT-UAV agriculture framework to provide a profound insight into the relationship between weather parameters and pet/disease occurrence. The UAV was flown based flight map, with adjusted dynamic speed and low altitude maintained to minimise power consumption. Additionally, the UAV was equipped with an integrated camera to transmit approximate images simultaneously, reduce the need to deploy sensor nodes in the farm field and compress by orders of magnitude. An open-weather application programming interface system and multiple LoRa were used to relay the collected data to the farmer's home. In contrast, TV White Space broadcasts the collected data from the farmer's house to IoT devices. The presence of diseases/pests in the crop can be detected using image processing of UAV-captured images. By comparing the climate and soil parameters after analysis, the framework shows higher predictive power for the occurrence of crop diseases/pests. Notably, the author introduced a novel approach to improve energy harvesting rates, increasing energy acquisition by 38% per day and introduced a flight mode. However, the system is limited to simulation.

Xu et al. [116] presented a novel study combining two different sensors mounted on a UAV to evaluate leaf nitrogen content in corn and explore the use of image information. A cornfield was divided into 48 plots, 24 of which were assigned to various cultivars. The others were related to nitrogen rates, each plot with an area of 6 × 10 m^2, where the data have been collected based UAV weighed 4.4 kg, a flight altitude of 60 m, speed of 6 m/s and 80% of front and side overlap and also compacted with two types of camera sensors to collect multispectral and digital RGB imagery for three corn stages including critical stage V12, period of transformation to reproductive R1 and during reproductive growth where the corn seeds being yellow R3. The high-density images were transferred into other spaces with different colours using hue-saturation-value to distinguish vegetation from soil, and a segment threshold was used to separate them. Notably, the RGB image transformation has been applied twice to facilitate distinguishing between soil and vegetation. Based on the demonstration studies, the random forest classification method was used to differentiate between soil and vegetation and generate a vector of them. On the other hand, the reflected bands of the multispectral images used a spectral canopy feature with variables related to canopy structure, plant activity and nutrient situation. Coverage-adjusted spectral and vegetation indices have been extracted from the multispectral images to indicate growth status and crop nutrition. Additionally, the partial-least-squares method was used to compare leaf-nitrogen content, and a set of coverage-adjusted spectral indices (vegetation indices) was selected using the random forest algorithm for corn growth stages. Coverage-adjusted spectral indices efficiently estimated leaf nitrogen content, outperforming vegetation indices across all corn growth stages. In contrast, at the R1 stage, coverage-adjusted spectral indices yielded 0.59, 22.02% and 8.37% for R^2, RMSE and NRMSE, respectively. However, the flight time was very short.

Yu et al. [116] presented a novel algorithm that combines regional-variance matching and intensity-hue-saturation fusion of visible and thermal infrared images captured by UAVs to enrich and improve information on wheat growth. For algorithm validation, high-definition digital photographs and thermal infrared imagery of two stages of wheat growth were captured using a miniaturised eight-rotor electrically driven UAV equipped with digital and thermal image cameras, simultaneously collecting imagery at 0.05 K resolution at a flight height of 50 m. The UAV-based digital photographs and thermal infrared images of wheat at two growth stages were used to validate the method. The general work-flowchart includes six steps: (i) After recording the visible and thermal infrared imagery of RBG space, they were transformed to intensity hue saturation space; (ii) the components configured of wavelet, select base and sym4, which represent the images intensities; (iii) high-frequency coefficients of low and high-frequency subbands were calculated to provide the regional variance based 3 $\times$ 3-pixel window as a weighted template; (iv) the inverse of wavelet transform and the fused image with new intensity were gained based on the regional variance of the fusion rules represented by the coefficients of low and high frequency and (v) finally, the fusion of visible and thermal infrared imagery was obtained by inverting the intensity-hue-saturation transform using the thermal infrared image's saturation, with the visible image containing the new intensity and hue components. The technology of image fusion has provided a visual accuracy understanding of a target image with increasing efficiency, and it is superior to intensity-hue-saturation transform and variance of the traditional region; thus, when identifying the wheat growth's stress characteristics, the fused images can permit more capabilities of interpretation than thermal infrared images or original visible due to it has high saturation and spatial resolution. However, this technology has a limitation regarding the UAV battery lifetime.

6.6 | Comparison of Previous Works-Based Drones in Agriculture

The previous section displays the different applications of drones in the agriculture sector in the last 5 years, which all have appearance multiadvantages by facilitating data collecting in a real-time manner, including thermal, RGB images and WSN fields by using different methodologies, hence enhancing the crop yield prediction, management and irrigation which is the promising problem in the future of the world. The survey included various types of UAVs, each characterised by its capabilities, which are compared in Table 4. The table compares the studies under review based on key parameters, including drone type, flight time, battery capacity and the size of the agricultural area. In addition, it highlights the limitations of individual studies.

Nonetheless, a limitation was identified in nearly all prior research on battery capacity, flight time and stability, which remains a primary challenge. At the same time, no one from the mentioned survey had a complete solution plan for such problems. The maximum flight time was 55 min in ref. [108] when using M300 RTK to collect a thermal image. In comparison, the minimum was 20 min in ref. [117] when using DJI-S1000+ to collect multispectral and digital RGB imagery, although the battery capacity was 10,000–20,000 mAh. In addition, ref. [109] used an MD4-100 Quadcopter to reduce power consumption during transmission, achieving a flight time of 45 min with a battery capacity of 17.3 Ah, which appears to be the most suitable of the surviving options.

7 | Discussion on Previous Works-Based WSNs and Drones

The importance of WSNs in agriculture is that they keep track of the surrounding environment and gather important data to enhance productivity and sustainability. They also watch ecological variables (temperature, humidity and solar radiation) and help farmers to make rational decisions—moisture in the soil can be monitored to maximise irrigation and avoid over- or under-watering. WSNs are also used in the management of fertilisers more efficiently. The information gathered is used to optimise the planting schedules and thus maximise the yields and reduce expenses. Unlike the traditional networks, which used simple sensor nodes to gather data, sensors are able to detect abnormal patterns in the plants and hence intervene in time before the spread of the pests or diseases.

Nonetheless, scalability and reliability are critical in WSNs, even though they are essential in providing real-time monitoring of agriculture. The majority of the available systems can only be tested on a small scale or controlled scenario and do not provide any assurance of consistent communication over large and heterogeneous farms under extreme environmental conditions. Sensor lifetime and even network capacity to satisfy energy needs are also constrained by energy requirements, and this demonstrates the importance of new energy-harvesting solutions and adaptive protocols to achieve longer operation. These challenges need to be met to implement resilient WSNs in large-scale and long-term agricultural systems. Hybrid solar and wind energy collection is a sustainable way of solving the battery constraints and maintenance challenges of remote farms today. Such systems can provide a continuous power supply in response to the changing weather and seasonal differences, improve the independence of nodes and the entire network stability by combining solar panels with microwind turbines. Also, smart power management circuits can be configured to conserve energy depending on sensor activity and environmental factors. This will be able to minimise the operational expenses and will allow constant checking of the soil, crop and

TABLE 4 | Comparison of previous agriculture studies using UAVs.

Ref./year	Drone type	Flight time (minutes)	Battery capacity (mAh)	Purpose of using UAV	Area size (m^2)	Limitations of UAV
Zhang et al. [105]/ 2020	DJI Matrice600pro	32	4500	To enable fast data collection in remote areas without public networks, extend the lifespan of the ground terminal. It has been demonstrated in remote environmental monitoring	N/A	The system's limited capacity of the upper communication link will lead to increased transmission times, and the received signal strength will decrease with increasing distance
CASTELLANOS et al. [109]/2020	MD4-100 Quadcopter	45	17,300	To collect data from potato crops	380 × 760	The system was limited to simulation
Heinemann et al. [112]/2020	Fixed-wing UAV	30–45	11.1 V; 10,000	To enable real-time capture of long-wave infrared and nearfield visible infrared as well as activating agricultural regions through semisupervised thermal data mapping	189 × 6	There was a limitation in the UAV's power supply
Gao et al. [115]/ 2020	DJI T600	30	4500	To collect images for plants and send them in real-time	9 × 3	Wind force and rain humidity also influence the emergence and spread of pests and diseases
Yu et al. [117]/ 2020	Miniaturised eight-rotor electricity-driven	20	5000	To collect visible and thermal infrared images simultaneously	N/A	Limited types of UAV images
Awais et al. [106]/ 2021	DJI Matrice 300 RTK	30–40	5935	Identify the best timing and altitude for high-res remote sensing thermography and calibrate leaves and canopy with an infrared thermometer	35 × 7	Limitation in UAV flight time
Xu et al. [107]/ 2021	DJI Matrice 100	25	4500	Integrate three cameras (RGB, multispectral and thermal) and a LiDAR sensor with software for data recording and visualisation	18 × 24	Limitation in UAV flight time
Awais et al. [108]/ 2021	M300 RTK	30 to 55	5935	To collect RGB and thermal images	27 × 7	There was a limitation in the UAV's power supply
Xu et al. [116]/ 2021	DJI-S1000+	20	10,000–20,000	To collect multispectral and digital RGB imagery	28.8 × 160	Limitation of flight time
Cariou et al. [118]/ 2022	Matrice 300 RTK—DJI	40	5935	To improve the feasibility of collecting data from underground sensors experimentally	100 × 10	There was a limitation in transmitting data from underground sensors to UAV (about 0.29–5 kbps) and the UAV battery lifetime
Yamagishi et al. [113]/2022	DJI Mavic Pro 2	31 min	3850 mAh	To collect RGB images	7 × 6	Limited stability
Sun et al. [114]/ 2024	DJI Mavic 2 Pro	31	3850	To collect RGB images	6 × 120	There was a limitation in the UAV's power supply and flight time

climate data, which will eventually lead to efficiency, sustainability and resilience in precision agriculture.

Agricultural WSN systems that are used today are increasingly incorporating hybrid solar-wind energy harvesting as a sustainable development strategy to cope with the problem of short battery life and maintain in isolated farmlands. These hybrid systems use the complementary power of solar energy and wind power, unlike conventional battery-powered nodes, which makes the power supply more reliable and consistent across daily and seasonal variations. Storage units are charged using solar panels throughout the day, and more energy is supplied by microwind turbines when there is low sunlight or at night. This two-source design has a great benefit in terms of energy supply, node autonomy and network strength, which reduces the human battery replacement and operation expenses. Intelligent power management circuits also make more efficient use of energy delivery according to the sensor activity, communications requirements and environmental factors. In PA, where the monitoring of soil moisture, crop health and microclimate should be continuous to provide sufficient irrigation and fertilisation, such hybrid systems are very beneficial. All in all, the combined approach of incorporating solar-wind energy systems enhances the efficiency and performance of agricultural WSNs and facilitates the sustainability and renewable energy objectives of smart farming.

Farmers apply UAVs in agriculture to monitor, gather data, address pests and diseases, locate cattle, spray fertilisers and irrigate their farms. They can be used for agricultural surveillance, but they have a number of limitations, such as limited flight time, limited battery capacity and they are also susceptible to bad weather. Their extensive usage is also limited by regulations. The coordination between UAVs, WSNs and IoT platforms should be improved to guarantee the effective and uninterrupted collection of data. The next generation of research must be aimed at enhancing the endurance of UAVs by designing energy-efficient or hybrid designs and designing automated data fusion algorithms to combine effectively aerial and ground sensor data. By solving these problems, UAVs can be further used in precision agriculture.

8 | IoT in Agriculture

IoT in agriculture is a new revolution in the traditional farming process as it links the devices and sensors to the internet. This technology allows real-time gathering, transmission and analysis of the data to offer farmers important knowledge to make more effective decisions [119]. IoT systems create a platform that increases agricultural processes and gradually turn farms into smart farms at various levels [120]. The data collected by the sensors in soil, crops and farm equipment includes real-time data of soil moisture, temperature, humidity, nutrient levels and weather conditions. The information allows for accurately scheduling the watering, fertilising and planting to minimise waste of resources. To explain, such sensors are employed by smart irrigation systems that utilise the sensors to provide the correct amount of water, thus reducing water waste [121].

Crop health management with IoT has significant importance as it helps detect diseases, pests, and nutrient deficiencies. State-of-the-art cameras, sensors and AI assess the health of their plants and assist farmers in making the necessary decisions in a timely manner in order to reduce losses [122]. Drones can now be used to survey a large field through aerial shots with the help of IoT sensors, thus making it possible to measure crop health better, predict yields and schedule harvesting time. In livestock production, IoT devices, such as smart collars, track the position of animals, their activities and temperature as well as their health. The data allow farmers to improve the welfare of animals, identify illness early and automate the process of milking and feeding, among other things, cutting down the costs and increasing the productivity of livestock [123].

IoT solutions allow controlling and monitoring all quality and safety aspects of food in the supply chain by taking into consideration all essential factors: temperature and humidity during transportation and storage [124]. This minimises wastage, enhances responsibility, boosts consumer confidence and minimises losses after harvesting. The IoT also has an interface with other areas of technology, including AI (including ML and DL and robotics), to provide self-sufficient smart systems in automated farms, including planting, spraying and harvesting [125]. The data obtained enable farmers to predict crop yields and manage input to achieve sales, enhance efficiency and profitability in farming. The IoT technology is also capable of solving other issues, such as climate change, labour crunch in labour-intensive activities and increasing population and food demand in the world [126]. Thus, IoT allows farmers to implement sustainable, scalable and space-saving agricultural practices to enhance innovation and resistance in farming.

PA involving the integration of WSNs, UAVs and IoT can be divided into three levels: data collection, data transmission and management, decision-making and control. All the stages assist in converting raw environmental data into usable agricultural information. WSNs and UAVs are collaborative in data collection, where they collect data at various places and on several occasions. Soil moisture, temperature, humidity and nutrient levels are continuously monitored by ground sensors, whereas aerial and multispectral imagery at a high resolution is taken by UAVs to determine the health of crops and their variability. This step gives comprehensive observation, although it is limited by sensor energy constraints, UAV flight time and problems in synchronising data sources [127]. The data transmission and management step takes advantage of the IoT networks and able to connect sensors, UAVs and cloud platforms, which are capable of monitoring large areas in real-time. However, latency, bandwidth and data loss (especially in rural areas) may also pose a problem. The most developed phase is the decision-support and control phase where the processed data are used to take decisions such as irrigation, fertilisation and pest control. This increases accuracy and efficiency in the resource use but depends on accuracy and reliability of previous data collection and transmission. In order to enhance these systems, it is important to identify important performance measures. Key metrics in the case of wireless sensor networks (WSNs) are energy consumption, network life, packet delivery ratio, latency and coverage [128]. The most important performance indices of

UAVs are the flight time, area coverage, image quality, data acquisition rate and productiveness of missions. Other parameters, including data transfer, stability of communication and processing delays, affect success. The awareness of such metrics assists in the development of mechanisms that maximise the use of energy, data accuracy and scalability. It allows the determination of the right technologies, depending on the environment and field size, data need and sustainability goals. In general, the classification and evaluation of such technologies would improve accuracy agriculture research and assist in resilient and energy sense-sensitive farming, based on data to address worldwide agricultural challenges [129].

Morchid et al. [130] presented an intelligent irrigation system by utilising the current developments in the embedded system, IoT and cloud-based technology. It actively supervises the parameters of the agricultural environment, which are humidity, soil moisture and temperature, using sensors linked to an ESP32 embedded platform. The water level in the irrigation tank is also checked with a water level sensor. The obtained data are sent via WiFi to the ThingSpeak cloud and can be accessed by the farm owners using the ThingSpeak app as well as automatically operate two water pumps. The system attains a water consumption of 70% reduction over the traditional water consumption methods. However, it is affected by a number of challenges, such as technological constraints, environmental effects, crop effects, climate change adaptation, agricultural education and management of the soil moisture.

Aziz et al. [131] developed a hybrid and self-configuring WSN-IoT-animal system that would facilitate the improvement of data gathering in vast farmlands and isolate PA in relation to traditional farming procedures. These sensors are distributed throughout the farm, and they collect environmental data and send it to a mobile cluster head, which is a sensor node attached to an animal. These sensors are selected on the basis of energy levels and distance to the target area. The data collected is then sent to the cloud by the cluster head and can be accessed by the farmers. The mobile cluster head maintained 97% of the sensors as opposed to 2000-round cluster heads. The efficiency and performance of large-scale data collection were improved by this approach, but this includes a rather limited range of communication and can be optimised in only a few different ways.

Ahmed et al. [132] introduced a portable tool, which used several sensors to collect soil information. It has humidity, pH, soil moisture and temperature sensors that were connected to an Arduino UNO that gathered and processed the data, and then sent it to the Blynk server to monitor the real-time parameters. Also, it sent SMS alerts to farmers using a GSM Module. The system had SMS notifications, along with live soil information, where the farmer can get valuable information immediately, thus avoiding a lot of irrigation due to measuring the pH of the soil. However, its functionality is restricted to Simulink.

Sadiq et al. [133] integrated IoT and various UAVs to track the health status of cattle in a dairy grazing system in real time. The animals were fitted with sensors to gather health data on the grazing grounds and relay it to UAVs. Besides this, UAVs were traversing to take pictures, connect them to the data gathered and subsequently transmit them to the cloud. Animal welfare and health information cloud-based web applications are available to the farmers. The system offered the benefits of improving the health and farm productivity of the animals. Nonetheless, this system was expensive and was not scalable, besides the fact that it had an underdeveloped technology infrastructure in the rural regions.

Wang et al. [134] created an initial step in the crop disease prevention model, which assists farmers in reducing the infection using DL models deployed in an agricultural IoT network. They came up by deploying a deep convolutional generative adversarial network. It was applied in the detection of bacterial leaf blight, the leaf streak and panicle blight on a satisfactory level. It showed better training stability, and it was able to work without segments when the data were preprocessed using augmentation, normalisation and singular value decomposition. The model was found to be 95.99% accurate on the Plant Village dataset, better than the traditional deep learning models at recognition and sample quality. Nonetheless, the flaws of this model were positive to a certain amount of operations in the agricultural domain. However, it was precise even when compared to human recognition.

Zhao et al. [135] suggested an IoT system that would support smart farming through big data-assisted geospatial analysis to enhance the harvest. The irrigation process and crop monitoring were done using a UAV. It used geographic information systems containing different applications, including site search, data analytics, land adaptability discovery, service allocation, land allocation, impact measurement and information systems. Data collection and real-time transmission to farmers were made possible with the use of IoT sensors. The prediction, yield, average error, accuracy and low-cost rate of 97.7, 92.4, 38.3, 94.5 and 34.4, respectively. However, the analysis was limited to simulation.

Shafi et al. [136] invented an all-in-one system of crop monitoring based on the latest UAV, IoT, and ML technologies. The data collection method encompasses various elements, such as IoT-agriculture nodes, to measure environmental conditions on crops and UAV platforms to supply multispectral data to do NDVI. The system addresses efficient integration and viable implementation. It was situated on a 4200 m^2 territory, where a grid 3×3 was prepared, and in each 0.15-acre cell, an IoT-Agri-node (LoRa, Raspberry Pi, a solar cell and battery) was installed to measure soil moisture, temperature and humidity data and transmit it to a master IoT node. This information is sent by the master node through GSM to a web portal. The communication failures are controlled by SD-RAM controller node. To classify the health of crops, SVM, naive Bayes and neural networks were compared and the M4 neural network was found to have the highest accuracy of 98.4. Nevertheless, the environmental factors remain problematic for the system performance grouping.

LU et al. [137] introduced a novel system of precision farming that uses the development technology constituents to help in the management and monitoring of vineyard health. UAV was employed in gathering data at various base stations of a WSN and transmitting these data to a cloud, where the users can access it remotely. The sensor nodes were not dependent upon each other and used unique IDs; a node sending collected data to the closest gateway base via ZigBee, and the UAV flew over the gateway and gathered and analysed data with the help of NDVI, and transmitted the data that was to be stored in the cloud. In 2019 and 2020, the experimental approach was practically implemented at the two sites, and it has led to better crop health because of the opportunity to schedule various cultivation and harvesting stages. Nevertheless, the resolution of the spatial images was 3 cm or less.

Almalki et al. [138] utilised underground sensors and above-ground sensors to keep track of the environment at any given time. The hourly data are dispatched to the ground gateways and uploaded to the cloud on a 12-h basis to be stored and analysed. The manufacturers, government entities and farmers can access this cost-effective system. It allowed real-time monitoring, which enabled prompt decision-making or a data-based decision on the present data. IoT sensors and UAVs are used to improve PA, which boosts crop production and saving of resources. Nevertheless, in the course of development and maintenance, some technical problems, such as connection problems, might occur, which will interfere with the transmission of data and the functioning of the system. Furthermore, the narrow scope and limits of sensors, gateways and drones affected the general performance of a system.

9 | Common Challenges and Limitations in Agriculture

Although digital agriculture has made great progress, the actual applications of WSNs, UAVs and IoT have been limited due to several technical and functional difficulties as Figure 3 demonstrates.

In terms of WSNs, challenges in agriculture can be concluded as follows:

1. Limited energy availability: WSNs generally use energy sources, such as batteries or energy-harvesting materials, which are not always capable of operating in remote areas on a long-term basis [139].
2. Communication issues: Wireless communication can be inefficient in extensive or distant regions because of interference, congested networks and the geographic location, resulting in loss of data [140]. Severe weather, such as rain, dust, temperature and physical destruction by animals or farm machinery, may affect sensor functions and longevity. Also, the signal quality may be compromised by the presence of other devices that will leave it susceptible to cyber-attacks, data manipulation or unauthorised access-which may undermine the accuracy and integrity of agricultural data collection. Therefore, path loss has been another major problem in WSNs in the agricultural industry because it reduces signal intensity as wireless signals are transmitted across the landscape [140–143].
3. Increased operational and maintenance costs: WSNs are expensive to deploy and maintain, especially in large farms where the need to increase the network coverage may necessitate heavy investment in hardware and manpower [142].
4. Limited bandwidth and data management analytics: WSNs have limited bandwidth, which can be used to transport a large amount of data, thus requiring compression or data processing. They also need an effective storage so that there is a certain degree of confidence in communication, which may be particularly difficult when it comes to real-time decision-making packages [144].
5. Sensor calibration and reliability: Continuous calibration and validation are necessary to ensure the accuracy and reliability of sensors with time, especially when the environmental conditions vary [145].

Moreover, smart agricultural systems have been promoted by the development of UAV technology. However, there are still a

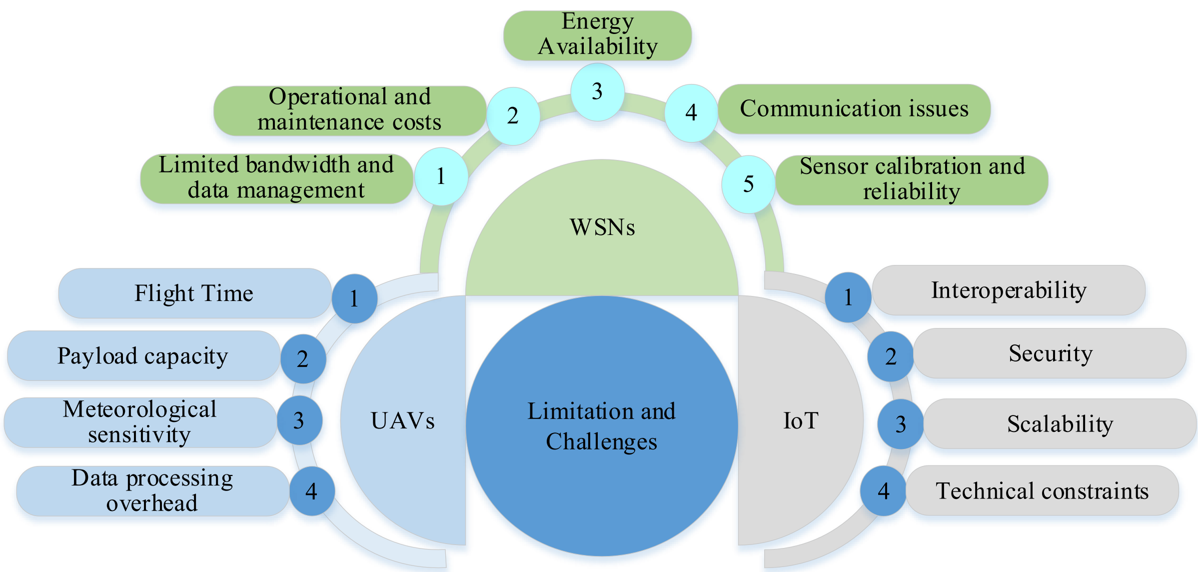

FIGURE 3 | WSNs, UAVs, IoT limitations and challenges in the agriculture field.

number of major obstacles, such as short flight time. Proper deployment of UAVs requires dependability, cost-effectiveness, compliance with regulations and efficient route planning. Quality services are necessary, and concerns, such as the use of LOS communications to communicate between the UAV and ground, dependence on global navigation satellite systems, and the use of constraints of size and weight are still troublesome [146–148].

Although these continue to face technical setbacks, such as aircraft stability, maintaining its altitude, weather sensitivity, automated takeoff and landing, capacity to carry payload and engine malfunction, these are the key issues to using UAVs in smart agriculture [149, 150]. Additionally, miniaturisation of sensors and platforms and camera performance, such as shadows and reflections in daytime, impacted imagery [151].

In the broader context of the IoT, significant challenges, such as device interoperability, data security and privacy, and scalable communication protocols, hamper integration in a system [152, 153]. The big data generated by IoT devices demand advanced analytics and cloud infrastructure, which can be expensive or difficult to implement in rural areas with limited connectivity [154]. Consequently, despite their potential to revolutionise PA, the technical limitations of these technologies, the required infrastructure and the regulatory issues should be properly discussed in order to realise sustainable and scalable solutions implementation.

10 | Future Trends in Agriculture

Agriculture is becoming digitalised with the combination of WSNs, UAVs, IoT and AI, and traditional farming is becoming more efficient, sustainable and data-driven. The following is the list of future agriculture trends in the area of these technologies.

10.1 | WSNs

WSNs will develop to have sensors to check on different environmental conditions such as water level, soil pH, air quality and temperature. These networks will be able to provide real-time information, which will help farmers to make more informed choices, and also automate procedures such as fertilisation, irrigation and pest management [155]. The next generation of WSNs will embrace the use of low-power and long-range communications, such as NB-IoT and LoRaWAN, in order to reach out to vast agricultural regions. This enables the farmers to manage remote areas and gather essential information without the concern of having to frequently replace their batteries or having poor connectivity [156].

10.2 | Drones

Precision agriculture relies on drones fitted with hyperspectral, multispectral and thermal sensors. They give a step-by-step aerial coverage and health of crops, which allows an early identification of ailments, deficiencies and pests. This sustains specific interventions, minimises the use of chemicals and enhances the efficiency of fertilisers and pesticides [157]. Automated drones will be able to spray weeds, plant seeds and even scout crop growth, thus cutting down on the amount of labour needed and boosting productivity [158]. Swarm technology has become a norm, and several drones can collaborate with ease. This assists farmers to work on a bigger area at a time and take up activities, such as irrigation, applying fertilisers and pest control, at the same time [159].

10.3 | IoT

The IoT will facilitate inter-agricultural systems where sensors, devices and equipment will be able to communicate with each other. As an illustration, irrigation systems enabled by the IoT can automatically regulate the water supply based on real-time data on the moisture levels of the soil. Similarly, weather conditions are observed using environmental sensors that are used by farmers to schedule the effective harvesting and planting time [160]. The IoT provides the ability to achieve continuous data recording of applications as geospatial sensors, weather stations or livestock monitoring systems. It is analysed and processed in real time and give farmers actionable insights to manage resources and crop health, and predict [161]. Automation by IoT, such as remote watering systems and automated tractors, will become more widespread, avoiding the use of human labour and enhancing the precision of current services, including irrigation, spraying and fertilisation [162].

10.4 | AI

AI is essential to the agricultural sector as it enables making data-driven decisions based on the data gathered by WSNs, drones and IoT gadgets. Machine learning models assist in detecting disease-causing bacteria in crops, predicting crop yields and prescribing the best farming methods to use, utilising both real-time and past data [162]. The optimisation of the PA methods by the AI algorithms will find the optimal conditions for different crops. They use data from various sources to find the optimal irrigation, plant planting schedules and fertiliser application and minimise waste and maximise yield [163]. The use of AI-powered robots will automate work that is labour-intensive, including harvesting, planting and weeding. These robots have the capability to learn the conditions in the field by machine learning, and they get better with time. As an example, robotic weed-controllers and autonomous weed harvesting systems eliminate the use of human labour and chemical herbicides. The integration of AIs will probably improve livestock management by using data on animal wearables, biosensors and other gadgets to track health, behaviour and location. Such a strategy may also facilitate the prevention of diseases early, the maximisation of feeding periods and improved treatment of animals [164].

A combination of AI and IoT is the key to PA development with the prior use of smart, independent and data-driven farm management. The IoT networks are made of sensors, actuators and other connected devices that measure different data about soil, crop health, microclimates and equipment operation [165]. The true benefit is the ability to know these data, which can be done effectively by AI with the help of ML, DL and knowledge reasoning. Through the combination of AI and IoT, raw sensor data can be converted into useful information and decisions can be made, including predicting the need to irrigate, forecasting pest outbreaks and analysing the optimal use of nutrients [166]. Edge AI and federated learning enable data processing near the source, minimising delays, bandwidth and cloud expenses, perfect for remote and low-power farming environments. The improvement of awareness by means of data fusion, anomaly detection and adaptive controls, which are achieved with a combination of AI and IoT, leads to high yield, resource utilisation and sustainability [167]. Nevertheless, this integration requires a strong standard of communication, interoperability and data governance to ensure reliability and privacy [168]. Altogether, the synthesis of AI and IoT is turning PA into a novel predictive system learning from its mistakes and self-evolving to enhance farming activities in various settings.

The autonomous systems using drones are a major step towards real-time data-driven crop management in the modern PA. These systems utilise aerial sensing, ML and edge computing to provide dynamic spatial insights into crop health, growth and environmental conditions [169]. Drones with high-quality sensors (including multispectral, hyperspectral and thermal imagers) are autonomous in nature and capable of continuously surveying key agronomic parameters, such as chlorophyll, canopy temperature, soil moisture and pests. Due to real-time image analysis and the implementation of AI-based anomaly detection, they will have the opportunity to detect the early signs of both biotic and abiotic stresses before they may manifest themselves and implement timely and specific interventions that will improve resource management and increase crop yields [170]. Furthermore, onboard decision-support algorithms enable drones to both sense and act autonomously, carrying out precision irrigation, localised spraying or nutrient distribution guided by in-flight analytics. This closed-loop system reduces reliance on human intervention, reduces waste and facilitates adaptive management amid evolving environmental conditions [171]. The combination of drone autonomy and ground sensors, and cloud IoT helps increase the amount of data collected at different places and at different intervals, which facilitates lifelong learning and construction of predictive models throughout crop cycles [128]. Therefore, an autonomous system drone is not a simple monitoring tool but an intelligent agent in a networked agricultural system, which has advanced to implement proactive, sustainable and self-optimising agricultural management.

The combination of AI, edge computing and digital twins is a crucial step to the creation of predictive and self-optimising methods in agriculture under Agriculture 5.0 [172]. These technologies generate a cyber-physical ecosystem where AI uses diverse agricultural data, edge computing is used to process low-latency data closer to the data source and digital twins represent a current virtual representation of the farming environment [173]. Field sensors, UAVs and autonomous machinery generate large real-time data streams that cover soil moisture, canopy temperature, nutrients and crop phenology, which are processed locally using edge intelligence to create immediate and actionable insights [166]. These insights update the digital twin, enabling AI models to run predictive simulations and improve irrigation, fertilisation, and pest control strategies. This decentralised intelligence system allows real-time adjustments and proactive decision-making, eliminating the need to rely on cloud-based systems and enhancing resilience in regions with poor connectivity [174]. Moreover, the cooperation between AI and digital twins facilitates continuous learning: AI improves its predictions as the results of the twin get better, and the latter adjusts in real-time to the existing conditions in the field. The joint methodology is very effective in enhancing accuracy, efficiency of operations and sustainable use of resources since it enables intervention of stressors at an early stage when they are not severe. However, developing fully mature AI–edge–digital twin systems based on progress in model resilience, interoperability and cyber-physical integration to turn predictive insights into practical agricultural outcomes [175].

New AI approaches (including generative, reinforcement and federated learning) are not based on reactive analysis but on the self-optimising systems capable of anticipating the changes in the environment and adjusting the management strategy on their own. Moreover, the miniaturisation of hardware and smarter edges will also allow the use of complex inference tasks that can be performed on farms, thus delivering almost instant feedback and avoiding the use of remote cloud servers [176]. It is also hoped that future digital twins will evolve into actively changing virtual ecosystems that constantly align themselves with real-time data through the feedback of AI. These advanced twins will replicate the activities, such as crop growth and soil moisture, besides giving more accurate predictions of pest explosions, nutrient shortages and yield variations. This will eventually result in complete data-driven autonomous agricultural systems, in which intelligent agents will operate in both physical and virtual spaces to improve sustainability, resiliency and productivity in the agricultural sector in the future [177].

10.5 | Integration of Technologies

Coupling of UAVs, WSNs and IoT has significantly increased agricultural productivity and reduced expenses because the data allow impartial management and resource optimisation. It has been shown that UAV multispectral imaging has been used to help identify stress and deficiencies in crops at an early stage, which results in an average 10%–20% yield increase when timely interventions are made [178]. Precision irrigation with the help of IoT, soil moisture sensors and wireless data reduces the water usage by 30%–40% without reducing or deteriorating the crop quality [179]. WSNs to monitor microclimate and soil condition have helped in minimising the use of fertilisers by 15%–25% as the nutrient delivery is targeted spatially, hence reducing the cost of inputs and also reducing the environmental impact [180]. Integrating UAV imagery with IoT analytics automates pest zone mapping and crop vigor, minimising manual scouting labour by 20%–60% [181]. Such integrated technologies have the potential to save 20%–35% of farm expenses, mainly through the optimisation of irrigation, predictive maintenance and accurate input application [182]. It is possible to note that this data-based methodology demonstrates the potential of the UAV-WSN-IoT combination to enhance the idea of sustainable intensification, which can advance economic and ecological performance in contemporary PA.

Nonetheless, there are still some significant challenges, especially in controlling energy and system integration. Each WSN node, UAV and IoT gateway has different energy demands, which complicates the coordination and long-term deployment. Sensor nodes are usually dependent on batteries or ad hoc energy collection, but UAVs are limited in terms of flight time, as they use a lot of energy [5]. This brings a big trade-off in terms of energy efficiency, data precision and timeliness. Moreover, transmitting raw data in high volumes through low-power wireless connections burns a lot of power and lowers the system's life. In order to address these challenges, scientists are currently focusing on cross-layer energy optimisation and are inventing sensing, processing and communication techniques to decrease energy consumption without compromising on service quality [183].

New solutions include top–down data processing and dynamic machine learning. The implementation of small ML models, referred to as tiny ML, can be directly embedded into sensors to facilitate the local processing of data, the identification of anomalies and event-based communication to minimise the amount of energy consumption during data transmission [184]. UAVs act as mobile edge computing nodes, whereby they receive summarised data using the ground nodes and perform initial processing before transmitting information to the cloud servers. In predictive energy management, the environmental prediction (solar radiation, temperature, etc.) is used and the sensors increase and decrease their sampling rate and duty cycles depending on the available energy. Also, federated and reinforcement learning techniques enable decentralised energy-efficient adaptation, aiding the network to optimise sensing and communication strategies with time. Model compression, quantisation and pruning are other useful techniques to lessen the computing burden, and lightweight security protocols that would not reduce the integrity of the data too much and increase the amount of energy usage are also used [185].

Sophisticated AI systems, including federated learning, reinforcement learning and deep reinforcement learning, are finding an increasing application to facilitate decentralised energy-efficient adaptation within WSN-UAV-IoT systems. The algorithms help devices to cooperate in order to optimise sensing, routing and transmission strategies, and ensure data privacy and minimal energy consumption. Also, the techniques of model compression, quantisation, pruning and knowledge distillation are useful in reducing the computational and memory resources, enabling AI models to run effectively on resource-constrained machines [186]. Efforts in the lightweight AI security protocols are also aimed at assuring data integrity and authentication as well as reducing energy expenditure.

On the whole, the integration of WSN, UAV, IoT and emerging AI technologies is one of the essential steps on the way to an emerging intelligent PA. By incorporating AI that is energy-conscious at every tier of the system, such as sensors, UAVs and cloud platforms, the agricultural network of the future will be more autonomous, flexible and resilient. Such synergy of energy-efficient design and built-in AI will be a great addition to the usability, scalability and reliability of precision farming, thus creating sustainable, data-driven and self-optimising agricultural ecosystems.

11 | Conclusion

Considering the unprecedented environmental changes, making sure that there is insurance of a sustainable food supply is paramount to the growing population in the world. The agricultural sector has to adapt its practice to ensure that issues, such as water resource management, pests, diseases, soil variability and climate change, are well handled with practical and data-driven methods to be able to sustain it. The changes have been transformed by recent technological developments, such as environmental monitoring and sensing, using WSNs, UAVs and IoT in smart agriculture. However, according to the articles reviewed, WSN is viewed as the root of an intelligent farming system. WSN protocol relies on the application requirements. In addition, special solutions are needed to address WSNs to reduce power usage and enhance radio propagation investigation to evaluate performance in both theory and practice. UAVs are critical in the solution of the conventional restrictions through the enhancement of data collection and analysis. IoT combined with WSN and UAVs is also a major step in the field of real-time monitoring of the environment, providing necessary information about soil quality and the conditions of crops and weather. Such systems, with the emerging AI, are able to provide critical solutions.

The combination of WSNs, UAVs, IoT and ML would provide unprecedented possibilities to improve resource management and attain higher crop yields. Farmers and policymakers should adopt specifically designed evidence-based measures to achieve these benefits. The farmers ought to use IoT-based WSNs to monitor soil moisture and microclimate continuously and make sure they calibrate and maintain sensors regularly to maintain the accuracy of data. The UAV imaging would be able to supplement the ground sensors with high-resolution real-time information to determine the health of crops, the presence of pests and irrigation efficiency. They are also supposed to use ML-based analytics platforms, which integrate the UAV and WSN information to produce predictive information about the best irrigation and fertiliser application, and the detection of diseases. The policymakers are encouraged to develop standardised frameworks of data interoperability, subsidise the use of sensors and UAVs and facilitate the training of programmers to increase the digital literacy and decision-making capabilities of farmers. Besides, the creation of open agricultural data platforms may contribute to the engagement between stakeholders and the achievement of the ability to make UAV-IoT-WSN-ML systems scalable, sustainable and fair in various agricultural settings.

Author Contributions

Nada M. Khalil Al-Ani: conceptualisation, data curation, formal analysis, funding acquisition, investigation, methodology, project administration, writing – original draft, writing – review and editing. **Sadik Kamel Gharghan:** data curation, formal analysis, supervision,

validation, writing – original draft. **Ziad Qais Al-Abbasi:** investigation, project administration, visualisation, writing – review and editing. **Hasan Kahtan:** funding acquisition, writing – review and editing.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

Data sharing is not applicable to this article as no datasets were generated or analysed during the current study.

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